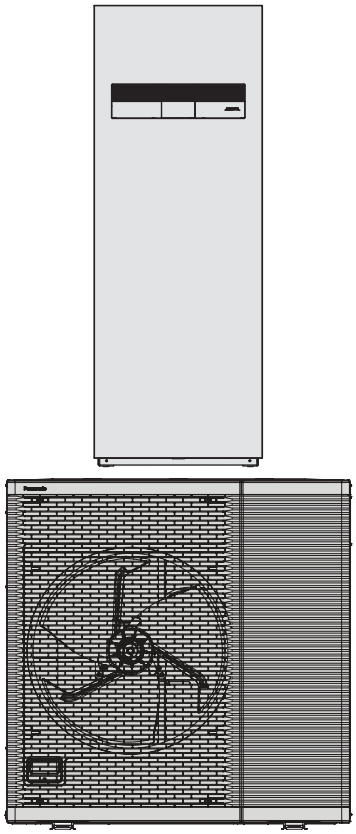
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Service Manual

Air-to-Water Hydromodule + Tank



Indoor Unit
WH-ADC0509L3E51
WH-ADC0509L3E5UK1

Outdoor Unit
WH-WDG05LE5
WH-WDG07LE5
WH-WDG09LE5

Destination
Europe
UK

⚠ WARNING

This service information is designed for experienced repair technicians only and is not designed for use by the general public.
It does not contain warnings or cautions to advise non-technical individuals of potential dangers in attempting to service a product.
Products powered by electricity should be serviced or repaired only by experienced professional technicians. Any attempt to service or repair the products dealt with in this service information by anyone else could result in serious injury or death.

IMPORTANT SAFETY NOTICE

There are special components used in this equipment which are important for safety. These parts are marked by ⚠ in the Schematic Diagrams, Circuit Board Diagrams, Exploded Views and Replacement Parts List. It is essential that these critical parts should be replaced with manufacturer's specified parts to prevent shock, fire or other hazards. Do not modify the original design without permission of manufacturer.

⚠ PRECAUTION OF LOW TEMPERATURE

In order to avoid frostbite, be assured of no refrigerant leakage during the installation or repairing of refrigerant circuit.

⚠ CAUTION

R290
REFRIGERANT

This AIR-TO-WATER HEATPUMP contains and operates with refrigerant R290.
THIS PRODUCT MUST ONLY BE INSTALLED OR SERVICED BY QUALIFIED PERSONNEL
Refer to National, State, Territory and local legislation, regulations, codes, installation & operation manuals, before the installation, maintenance and/or service of this product.

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• Specifications, designs and contents in this Service Manual are subject to change without notice.

1. Safety Precautions

- Read the following “SAFETY PRECAUTIONS” carefully before installation of Air-To-Water Hydromodule + Tank (here after referred to as “Tank Unit”).
- Electrical works and water installation works must be done by licensed electrician and licensed water system installer respectively. Be sure to use the correct rating and main circuit for the model to be installed.
- The caution items stated here must be followed because these important contents are related to safety. The meaning of each indication used is as below.
  Incorrect installation due to ignorance or negligence of the instructions will cause harm or damage, and the seriousness is classified by the following indications.
- Please leave this installation manual with the unit after installation.

⚠ **WARNING**	This indication shows the possibility of causing death or serious injury.
⚠ **CAUTION**	This indication shows the possibility of causing injury or damage to properties only.

The items to be followed are classified by the symbols:

⃠	Symbol with white background denotes item that is PROHIBITED.
❗ ⏚	Symbol with dark background denotes item that must be carried out.

- Carry out test run to confirm that no abnormality occurs after the installation. Then, explain to user the operation, care and maintenance as stated in instructions. Please remind the customer to keep the operating instructions for future reference.
- This appliance is not intended for accessibility by the general public.
- If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.

	⚠ **WARNING**	
1.	Do not use means to accelerate the defrosting process or to clean, other than those recommended by the manufacturer. Any unfit method or using incompatible material may cause product damage, burst and serious injury.	⃠
2.	Do not install outdoor unit near handrail of veranda. When installing outdoor unit at veranda of high rise building, child may climb up to outdoor unit and cross over the handrail and causing accident.	⃠
3.	Do not use unspecified cord, modified cord, joint cord or extension cord for power supply cord. Do not share the single outlet with other electrical appliances. Poor contact, poor insulation or over current will cause electrical shock or fire.	⃠
4.	Do not tie up the power supply cord into a bundle by band. Abnormal temperature rise on power supply cord may happen.	⃠
5.	Do not insert your fingers or other objects into the unit, high speed rotating fan may cause injury. 	⃠
6.	Do not sit or step on the unit, you may fall down accidentally. 	⃠
7.	Keep plastic bag (packaging material) away from small children, it may cling to nose and mouth and prevent breathing.	⃠
8.	Do not purchase unauthorized electrical parts for installation, service, maintenance and etc.. They might cause electrical shock or fire.	⃠
9.	Do not modify the wiring of outdoor unit for installation of other components (i.e. heater, etc). Overloaded wiring or wire connection points may cause electrical shock or fire.	⃠
10.	Do not pierce or burn as the appliance is pressurized. Do not expose the appliance to heat, flame, sparks, or other sources of ignition. Else, it may explode and cause injury or death.	⃠
11.	Do not pierce or burn as the appliance is pressurized. Do not expose the appliance to heat above 370°C, flame, sparks, or other sources of ignition. Else, it may explode and cause injury or death.	⃠
12.	Do not add or replace refrigerant other than specified type. It may cause product damage, burst and injury etc.	⃠
13.	Do not place containers with liquids on top of the Tank Unit. It may cause Tank Unit damage and/or fire could occurs if they leak or spill onto the Tank Unit.	⃠
14.	Do not use joint cable for Tank Unit / Outdoor Unit connection cable. Use specified Tank Unit / Outdoor Unit connection cable, refer to instruction 4 **CONNECT THE CABLE TO THE TANK UNIT** and connect tightly for Tank Unit / Outdoor Unit connection. Clamp the cable so that no external force will be acted on the terminal. If connection or fixing is not perfect, it will cause heat up or fire at the connection.	⃠
15.	For electrical work, follow the national regulation, legislation and this installation instructions. An independent circuit and single outlet must be used. If electrical circuit capacity is not enough or defect found in the electrical work, it will cause electrical shock or fire.	❗

	⚠ WARNING	
16.	For electrical work, follow local wiring standard, regulation and this installation instruction. An independent circuit and single outlet must be used. If electrical circuit capacity is not enough or defect found in electrical work, it will cause electrical shock or fire.	
17.	For water circuit installation work, follow to relevant European and national regulations (including EN61770) and local plumbing and building regulation codes.	
18.	Engage authorized dealer or specialist for installation. If installation done by the user is incorrect, it will cause water leakage, electrical shock or fire.	
19.	Engage dealer or specialist for installation. If installation done by the user is defective, it will cause water leakage, electrical shock or fire.	
20.	• The refrigerant cycle is completed inside the outdoor unit. • Refrigerant piping work is not required. • Pump down operation is not also required.	
21.	Install at a strong and firm location which is able to withstand weight of the set. If the strength is not enough or installation is not properly done, the set will drop and cause injury.	
22.	Install at a strong and firm location which is able to withstand the set's weight. If the strength is not enough or installation is not properly done, the set will drop and cause injury.	
23.	This equipment is strongly recommended to be installed with Residual Current Device (RCD) on-site according to the respective national wiring rules or country–specific safety measures in terms of residual current.	
24.	For refrigeration system work, install according to this installation instructions strictly. If installation is defective, it will cause water leakage, electrical shock or fire.	
25.	Do not use joint cable for outdoor connection cable. Use specified outdoor connection cable, refer to instruction ④ **CONNECT THE CABLE TO THE OUTDOOR UNIT** and connect tightly for outdoor connection. Clamp the cable so that no external force will be acted on the terminal. If connection or fixing is not perfect, it will cause heat up or fire at the connection.	
26.	Wire routing must be properly arranged so that control board cover is fixed properly. If control board cover is not fixed perfectly, it will cause fire or electrical shock.	
27.	After completion of installation, confirm there is no leakage of refrigerant gas. It may lead to the risk of fire or explosion when the refrigerant contacts with fire.	
28.	Ventilate the room if there is refrigerant gas leakage during operation. Extinguish all fire sources if present. It may lead to the risk of fire or explosion when the refrigerant contacts with fire.	
29.	Use the attached accessories parts and specified parts for installation. Otherwise, it will cause the set to fall, water leakage, fire or electrical shock.	
30.	Only use the supplied or specified installation parts. Else, it may causes unit vibrate, fall, water leakage, electrical shock or fire.	
31.	Only use the supplied or specified installation parts, else, it may cause unit vibrate loose, water leakage, electrical shock or fire.	
32.	If there is any doubt about the installation procedure or operation, always contact the authorized dealer for advice and information.	
33.	Select a location where in case of water leakage, the leakage will not cause damage to other properties.	
34.	When installing electrical equipment at wooden building of metal lath or wire lath, in accordance with electrical facility standard, no electrical contact between equipment and building is allowed. Insulator must be installed in between.	
35.	Any work carried out on the Tank Unit after removing any panels which is secured by screws, must be carried out under the supervision of authorized dealer and licensed installation contractor.	
36.	Any work carried out on the outdoor unit after removing any panels which is secured by screws, must be carried out under the supervision of authorized dealer and licensed installation contractor.	
37.	This system is multi supply appliance. All circuits must be disconnected before accessing the unit terminals.	
38.	For cold water supply has a backflow regulator, check valve or water meter with check valve, provisions for thermal expansion of water in the hot water system must be provided. Otherwise it will cause water leakage.	
39.	The piping installation work must be flushed before Tank Unit is connected to remove contaminants. Contaminants may damage the Tank Unit components.	
40.	This installation may be subjected to building regulation approval applicable to respective country that may require to notify the local authority before installation.	
41.	The Tank Unit must be shipped and stored in upright condition and dry environment. It may laid on its back when being moved into the building.	
42.	Work done to the Tank Unit after remove the front plate cover that secured by screws, must be carried out under the supervision of authorized dealer, licensed installation contractor, skilled person and instructed person.	
43.	Be aware that refrigerants may not contain an odour.	
44.	This equipment must be properly earthed. Earth line must not be connected to gas pipe, water pipe, earth of lightning rod and telephone. Otherwise, it may cause electrical shock in case of equipment breakdown or insulation breakdown.	
45.	This unit must be properly earthed. The electrical earth must not be connected to a gas pipe, water pipe, the earth of lightening rod or a telephone. Otherwise there is a danger of electrical shock in the event of an insulation breakdown or electrical earth fault in the outdoor unit.	

	⚠ CAUTION
1.	Do not install the Tank Unit at place where leakage of flammable gas may occur. In case gas leaks and accumulates at surrounding of the unit, it may cause fire.
2.	Do not install the outdoor unit at place where leakage of flammable gas may occur. In case gas leaks and accumulates at surrounding of the unit, it may cause fire.
3.	Prevent liquid or vapor from entering sumps or sewers since vapor is heavier than air and may form suffocating atmospheres.
4.	Do not release refrigerant during repairing a refrigeration parts. Take care of the liquid refrigerant, it may cause frostbite.
5.	Do not install this appliance in a laundry room or other high humidity location. This condition will cause rust and damage to the unit.
6.	Make sure the insulation of power supply cord does not contact hot part (i.e. water piping) to prevent from insulation failure (melt).
7.	Make sure the insulation of power supply cord does not contact hot part (i.e. refrigerant piping) to prevent from insulation failure (melt).
8.	Do not touch the sharp aluminium fin, sharp parts may cause injury.
9.	Do not apply excessive force to water pipes that may damage the pipes. If water leakage occurs, it will cause flooding and damage to other properties.
10.	Do not transport the Tank Unit with water inside the unit. It may cause damage to the unit.
11.	Carry out drainage piping as mentioned in installation instructions. If drainage is not perfect, water may enter the room and damage the furniture.
12.	Select an installation location which is easy for maintenance. Incorrect installation, service or repair of this Tank Unit may increase the risk of rupture and this may result in loss damage or injury and/or property.
13.	Select an installation location which is easy for maintenance. Incorrect installation, service or repair of this outdoor unit may increase the risk of rupture and this may result in loss damage or injury and/or property.
14.	Power supply connection to Tank Unit. • Power supply point should be in easily accessible place for power disconnection in case of emergency. • Must follow local national wiring standard, regulation and this installation instruction. • Strongly recommended to make permanent connection to a circuit breaker. - Power Supply 1: Use approved 25A 2-poles circuit breaker with a minimum contact gap of 3.0mm. - Power Supply 2: Use approved 15/16A 2-poles circuit breaker with a minimum contact gap of 3.0mm.
15.	Ensure the correct polarity is maintained throughout all wiring. Otherwise, it will cause electrical shock or fire.
16.	After installation, check the water leakage condition in connection area during test run. If leakage occurs, it will cause damage to other properties.
17.	If the Tank Unit not operates for long time, the water inside the Tank Unit should be drained.
18.	Installation work. It may need three or more people to carry out the installation work. The weight of Tank Unit might cause injury if carried by one person.
19.	Installation work. It may need two or more people to carry out the installation work. The weight of outdoor unit might cause injury if carried by one person.
20.	Keep any required ventilation openings clear of obstruction.
21.	Water piping in the occupied space shall be installed in such a way to protect against accidental damage in operation and service.
22.	Precautions shall be taken to avoid excessive vibration or pulsation to water piping.
23.	Protect the water piping from accidental rupture due to moving furniture or reconstruction activities.
24.	• Must ensure the installation of water pipe-work shall be kept to a minimum. Avoid use dented pipe and do not allow acute bending. • Must ensure that water pipe-work shall be protected from physical damage.

2. Precaution for Using R290 Refrigerant

- Pay careful attention to the following points:

	⚠ WARNING	
1.	The mixing of different refrigerants within a system is prohibited.	❗
2.	Operation, maintenance, repairing and refrigerant recovery should be carried out by trained and certified personnel in the use of flammable refrigerants and as recommended by the manufacturer. Any personnel conducting an operation, servicing or maintenance on a system or associated parts of the equipment should be trained and certified.	❗
3.	Any part of refrigerating circuit (evaporators, air coolers, AHU, condensers or liquid receivers) or piping should not be located in the proximity of heat sources, open flames, operating gas appliance or an operating electric heater.	❗
4.	The user/owner or their authorized representative shall regularly check the alarms, mechanical ventilation and detectors, at least once a year, where as required by national regulations, to ensure their correct functioning.	❗
5.	A logbook shall be maintained. The results of these checks shall be recorded in the logbook.	❗
6.	In case of ventilations in occupied spaces shall be checked to confirm no obstruction.	❗
7.	Before a new refrigerating system is put into service, the person responsible for placing the system in operation should ensure that trained and certified operating personnel are instructed on the basis of the instruction manual about the construction, supervision, operation and maintenance of the refrigerating system, as well as the safety measures to be observed, and the properties and handling of the refrigerant used.	❗
8.	The general requirement of trained and certified personnel are indicated as below: a) Knowledge of legislation, regulations and standards relating to flammable refrigerants; and, b) Detailed knowledge of and skills in handling flammable refrigerants, personal protective equipment, refrigerant leakage prevention, handling of cylinders, charging, leak detection, recovery and disposal; and, c) Able to understand and to apply in practice the requirements in the national legislation, regulations and Standards; and, d) Continuously undergo regular and further training to maintain this expertise.	❗
9.	Ensure protection devices, refrigerating circuit and fittings are well protected against adverse environmental effects (such as the danger of water collecting and freezing in relief pipes or the accumulation of dirt and debris).	❗

⚠ CAUTION

1.

Installation (Space)

- Must comply with national gas regulations, state municipal rules and legislation. Notify relevant authorities in accordance with all applicable regulations.
- Must ensure mechanical connections be accessible for maintenance purposes.
- In cases that require mechanical ventilation, ventilation openings shall be kept clear of obstruction.
- When disposal of the product, do follow to the precautions in #12 and comply with national regulations.
- Always contact to local municipal offices for proper handling.

2.

Servicing

2-1. Service personnel

- Any qualified person who is involved with working on or breaking into a refrigerant circuit should hold a current valid certificate from an industry-accredited assessment authority, which authorizes their competence to handle refrigerants safely in accordance with an industry recognized assessment specification.
- Servicing shall only be performed as recommended by the equipment manufacturer. Maintenance and repair requiring the assistance of other skilled personnel shall be carried out under the supervision of the person competent in the use of flammable refrigerants.
- Servicing shall be performed only as recommended by the manufacturer.
- The system is inspected, regularly supervised and maintained by a trained and certified service personnel who is employed by the person user or party responsible.
- Ensure refrigerant charge not to leak.

2-2. Work

- Prior to beginning work on systems containing flammable refrigerants, safety checks are necessary to ensure that the risk of ignition is minimised.
  For repair to the refrigerating system, the precautions in #2-2 to #2-8 must be followed before conducting work on the system.
- Work shall be undertaken under a controlled procedure so as to minimize the risk of a flammable gas or vapour being present while the work is being performed.
- All maintenance staff and others working in the local area shall be instructed and supervised on the nature of work being carried out.
- Avoid working in confined spaces. Always ensure away from source, at least 2 meter of safety distance, or zoning of free space area of at least 2 meter in radius.
- Wear appropriate protective equipment, including respiratory protection, as conditions warrant.
- Keep all sources of ignition and hot metal surfaces away.

2-3. Checking for presence of refrigerant

- The area shall be checked with an appropriate refrigerant detector prior to and during work, to ensure the technician is aware of potentially flammable atmospheres.
- Ensure that the leak detection equipment being used is suitable for use with flammable refrigerants, i.e. non sparking, adequately sealed or intrinsically safe.
- In case of leakage/spillage happened, immediately ventilate area and stay upwind and away from spill/release.
- In case of leakage/spillage happened, do notify persons down wind of the leaking/spill, isolate immediate hazard area and keep unauthorized personnel out.

2-4. Presence of fire extinguisher

- If any hot work is to be conducted on the refrigerating equipment or any associated parts, appropriate fire extinguishing equipment shall be available at hand.
- Have a dry powder or CO_2 fire extinguisher adjacent to the charging area.

2-5. No ignition sources

- No person carrying out work in relation to a refrigerating system shall use any sources of ignition in such a manner that it may lead to the risk of fire or explosion. He/She must not be smoking when carrying out such work.
- All possible ignition sources, including cigarette smoking, should be kept sufficiently far away from the site of installation, repairing, removing and disposal, during which flammable refrigerant can possibly be released to the surrounding space.
- Prior to work taking place, the area around the equipment is to be surveyed to make sure that there are no flammable hazards or ignition risks.
- “No Smoking” signs shall be displayed.

2-6. Ventilated area

- Ensure that the area is in the open or that it is adequately ventilated before breaking into the system or conducting any hot work.
- A degree of ventilation shall continue during the period that the work is carried out.
- The ventilation should safely disperse any released refrigerant and preferably expel it externally into the atmosphere.

⚠ CAUTION

2-7. Checks to the refrigerating equipment

- Where electrical components are being changed, they shall be fit for the purpose and to the correct specification.
- At all times the manufacturer's maintenance and service guidelines shall be followed.
- If in doubt consult the manufacturer's technical department for assistance.
- The following checks shall be applied to installations using flammable refrigerants.
  - The ventilation machinery and outlets are operating adequately and are not obstructed.
  - If an indirect refrigerating circuit is being used, the secondary circuit shall be checked for the presence of refrigerant.
  - Marking to the equipment continues to be visible and legible. Markings and signs that are illegible shall be corrected.
  - Refrigerating pipe or components are installed in a position where they are unlikely to be exposed to any substance which may corrode refrigerant containing components, unless the components are constructed of materials which are inherently resistant to being corroded or are properly protected against being so corroded.

2. 2-8. Checks to electrical devices

- Repair and maintenance to electrical components shall include initial safety checks and component inspection procedures.
- Initial safety checks shall include but not limit to:-
  - That capacitors are discharged: this shall be done in a safe manner to avoid possibility of sparking.
  - That there are no live electrical components and wiring are exposed while charging, recovering or purging the system.
  - That there is continuity of earth bonding.
- At all times the manufacturer's maintenance and service guidelines shall be followed.
- If in doubt consult the manufacturer's technical department for assistance.
- If a fault exists that could compromise safety, then no electrical supply shall be connected to the circuit until it is satisfactorily dealt with.
- If the fault cannot be corrected immediately but it is necessary to continue operation, an adequate temporary solution shall be used.
- The owner of the equipment must be informed or reported so all parties are advised thereinafter.

Repairs to sealed components

3.

- During repairs to sealed components, all electrical supplies shall be disconnected from the equipment being worked upon prior to any removal of sealed covers, etc.
- If it is absolutely necessary to have an electrical supply to equipment during servicing, then a permanently operating form of leak detection shall be located at the most critical point to warn of a potentially hazardous situation.
- Particular attention shall be paid to the following to ensure that by working on electrical components, the casing is not altered in such a way that the level of protection is affected. This shall include damage to cables, excessive number of connections, terminals not made to original specification, damage to seals, incorrect fitting of glands, etc.
- Ensure that apparatus is mounted securely.
- Ensure that seals or sealing materials have not degraded such that they no longer serve the purpose of preventing the ingress of flammable atmospheres.
- Replacement parts shall be in accordance with the manufacturer's specifications.

NOTE:	The use of silicon sealant may inhibit the effectiveness of some types of leak detection equipment. Intrinsically safe components do not have to be isolated prior to working on them.

Repair to intrinsically safe components

4.

- Do not apply any permanent inductive or capacitance loads to the circuit without ensuring that this will not exceed the permissible voltage and current permitted for the equipment in use.
- Intrinsically safe components are the only types that can be worked on while live in the presence of a flammable atmosphere.
- The test apparatus shall be at the correct rating.
- Replace components only with parts specified by the manufacturer. Unspecified parts by manufacturer may result ignition of refrigerant in the atmosphere from a leak.

Cabling

5.

- Check that cabling will not be subject to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects.
- The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.

Detection of flammable refrigerants

6.

- Under no circumstances shall potential sources of ignition be used in the searching or detection of refrigerant leaks.
- A halide torch (or any other detector using a naked flame) shall not be used.

⚠ CAUTION

7.

The following leak detection methods are deemed acceptable for all refrigerant systems.

- No leaks shall be detected when using detection equipment with a sensitivity of 5 grams per year of refrigerant or better under a pressure of at least 0,25 times the maximum allowable pressure (>0.98MPa, max 3.90MPa). For example, a universal sniffer.
- Electronic leak detectors may be used to detect flammable refrigerants, but the sensitivity may not be adequate, or may need re-calibration.
  (Detection equipment shall be calibrated in a refrigerant-free area.)
- Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant used.
- Leak detection equipment shall be set at a percentage of the LFL of the refrigerant and shall be calibrated to the refrigerant employed and the appropriate percentage of gas (25 % maximum) is confirmed.
- Leak detection fluids are also suitable for use with most refrigerants, for example, bubble method and fluorescent method agents. The use of detergents containing chlorine shall be avoided as the chlorine may react with the refrigerant and corrode the copper pipe-work.
- If a leak is suspected, all ignition sources shall be removed/extinguished.
- If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system. The precautions in #8 must be followed to remove the refrigerant.



8.

Removal and evacuation

- When breaking into the refrigerant circuit to make repairs – or for any other purpose – conventional procedures shall be used.
  However, it is important that best practice is followed since flammability is a consideration.
  The following procedure shall be adhered to:

> • remove refrigerant -> • purge the circuit with inert gas -> • evacuate -> • purge with inert gas ->
> • open the circuit by cutting.
> Brazing must not be used.

- The refrigerant charge shall be recovered into the correct recovery cylinders.
- The system shall be purged with OFN to render the appliance safe. (remark: OFN = oxygen free nitrogen, type of inert gas)
- This process may need to be repeated several times.
- Compressed air or oxygen shall not be used for this task.
- Purging shall be achieved by breaking the vacuum in the system with OFN and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum.
- This process shall be repeated until no refrigerant is within the system. (Until the concentration of purge gas is 0.25 LFL or less by the leak detector). ※0.25LFL = 0.525Vol%
- When the final OFN charge is used, the system shall be vented down to atmospheric pressure to enable work to take place.
- This operation is absolutely vital if brazing operations on the pipe work are to take place.
- Ensure that the outlet for the vacuum pump is not close to any potential ignition sources and there is ventilation available.



9.

Charging procedures

- In addition to conventional charging procedures, the following requirements shall be followed.
  - Ensure that contamination of different refrigerants does not occur when using charging equipment.
  - Hoses or lines shall be as short as possible to minimize the amount of refrigerant contained in them.
  - Cylinders shall be kept in an appropriate position according to the instructions.
  - Ensure that the refrigerating system is earthed prior to charging the system with refrigerant.
  - Label the system when charging is complete (if not already).
  - Extreme care shall be taken not to over fill the refrigerating system.
- Prior to recharging the system it shall be pressure tested with OFN (refer to #8).
- The system shall be leak tested on completion of charging but prior to commissioning.
- A follow up leak test shall be carried out prior to leaving the site.
- Electrostatic charge may accumulate and create a hazardous condition when charging and discharging the refrigerant. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.



⚠ CAUTION

Decommissioning

- Before carrying out this procedure, it is essential that the technician is completely familiar with the equipment and all its details.
- It is recommended good practice that all refrigerants are recovered safely.
- Re-use of recovered refrigerant is prohibited.
- It is essential that electrical power is available before the task is commenced.
  a) Become familiar with the equipment and its operation.
  b) Isolate system electrically.
  c) Before attempting the procedure ensure that:

10.

> • mechanical handling equipment is available, if required, for handling refrigerant cylinders;
> • all personal protective equipment and leak detectors are available and being used correctly;
> • the recovery process is supervised at all times by a competent person;
> • recovery equipment and cylinders conform to the appropriate standards.

  d) Make sure that cylinder is situated on the scales before recovery takes place.
  e) Start the recovery machine and operate in accordance with instructions.
  f) Do not over fill cylinders. (No more than 80 % volume liquid charge).
  g) Do not exceed the maximum working pressure of the cylinder, even temporarily.
  h) When the cylinders have been filled correctly and the process completed, make sure that the cylinders and the equipment are removed from site promptly and all isolation valves on the equipment are closed off.
- Electrostatic charge may accumulate and create a hazardous condition when charging or discharging the refrigerant. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before charging/discharging.

Labelling

11.

- Equipment shall be labelled stating that it has been de-commissioned and emptied of refrigerant.
- The label shall be dated and signed.
- Ensure that there are labels on the equipment stating the equipment contains flammable refrigerant.

Recovery

- When removing refrigerant from a system, either for servicing or decommissioning, it is recommended good practice that all refrigerants are removed safely.
- When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed.
- Ensure that the correct number of cylinders for holding the total system charge are available.
- All cylinders to be used are designated for the recovered refrigerant and labelled for that refrigerant (i.e. special cylinders for the recovery of refrigerant).
- Cylinders shall be complete with pressure relief valve and associated shut-off valves in good working order.
- Recovery cylinders are evacuated and, if possible, cooled before recovery occurs.
- The recovery equipment shall be in good working order with a set of instructions concerning the equipment that is at hand and shall be suitable for the recovery of flammable refrigerants.
- Make sure the recovery equipment is not a potential ignition source and is suitable for the refrigerant you are using.

12.

- In addition, a set of calibrated weighing scales shall be available and in good working order.
- Hoses shall be complete with leak-free disconnect couplings and in good condition.
- Before using the recovery machine, check that it is in satisfactory working order, has been properly maintained and that any associated electrical components are sealed to prevent ignition in the event of a refrigerant release. Consult manufacturer if in doubt.
- The recovered refrigerant shall be returned to the refrigerant supplier in the correct recovery cylinder, and the relevant Waste Transfer Note arranged.
- Do not mix refrigerants in recovery units and especially not in cylinders.
- If compressors or compressor oils are to be removed, ensure that they have been evacuated to an acceptable level to make certain that flammable refrigerant does not remain within the lubricant.
- The evacuation process shall be carried out prior to returning the compressor to the suppliers.
- Only electric heating to the compressor body shall be employed to accelerate this process.
- When oil is drained from a system, it shall be carried out safely.

3. Specifications

3.1 WH-ADC0509L3E51 WH-WDG05LE5

Item		Unit	Outdoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Cooling Capacity		Condition (Ambient/Water)	A35W7		
		kW	5.00		
		BTU/h	17100		
Cooling EER		W/W	3.23		
Heating Capacity		Condition (Ambient/Water)	A7W35	A2W35	
		kW	5.00	5.00	
		BTU/h	17100	17100	
Heating COP		W/W	5.05	3.52	
Heating Erp	Low Temperature Application (W35)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	5.0	5.0	6.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	6.00 / 237	5.06 / 200	4.25 / 167
	Annual Consumption	kWh	1113	2040	3483
	Class		A+++	A+++	A++
	Medium Temperature Application (W55)		Warmer	Average	Colder
	Application	Climate			
	Pdesign	kW	5.0	5.0	6.0
	Tbivalent / TOL	°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns	(W/W) / %	4.27 / 168	3.63 / 142	3.28 / 128
	Annual Consumption	kWh	1565	2849	4516
	Class		A+++	A++	A++
	DHW		Warmer	Average	Colder
	Application	Climate			
	COP / nwh	(W/W) / %	3.35 / 134	2.88 / 115	2.26 / 90
	AEC	kWh	384	447	569
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: -	Heating: -	Heating: -
		Power Level dB	Cooling: 61***	Heating: 57*** Heating: 52***	Heating: 57*** Heating: 52***
Air Flow		m^3/min (ft^3/min)	Cooling: 55.0 (1942) Heating: 45.0 (1589)		
Refrigeration Control Device			Expansion Valve		
Refrigeration Oil		cm^3	PZ68S (1100)		
Refrigerant		kg (oz)	R290, 0.96 (33.9) (Pre-charged) (-) (Maximum)		
F-GAS	GWP		3		
	CO^2eq (ton) (Precharged / Maximum)		0.003 / -		
Dimension	Height	mm (inch)	996 (39-7/32)		
	Width	mm (inch)	980 (38-37/64)		
	Depth	mm (inch)	430 (16-59/64)		
Net Weight		kg (lbs)	98 (216)		

Item		Unit	Outdoor Unit		
Pipe Diameter (Inner)		mm	20		
Standard Length		m (ft)	5.0 (16.4)		
Maximum Pipe Length		m (ft)	30.0 (98.4)		
I/D & O/D Height Difference		m (ft)	10 (32.8)		
Water Pipe Connector	Indoor	inch	1		
	Outdoor		1		
Compressor	Type		Hermetic Motor Compressor (Rotary)		
	Motor Type		Synchronous Electric Motor (4-poles)		
	Rated Output	kW	1.70		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	kW	1.089 (Heating) / 1.705 (Cooling)		
	Output Power	W	120		
	Fan Speed	rpm	Cooling: 520 Heating: 440		
Heat Exchanger	Fin material		Aluminium (Blue Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 46 × 19		
	Size (W × H × L)	mm	36.38 × 966 × 880.6 : 908.6		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		26		
	Size (W x H x L)	mm	57.8 × 524 × 117		
	Water Flow Rate	l/min (m^3/h)	Cooling: 14.3 (0.9) Heating: 14.3 (0.9)		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 1.55	Heating: 0.99	Heating: 1.42
Maximum Input Power For Heatpump System		kW	2.93		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 2.93k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	6.9		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 6.9	Heating: 4.4	Heating: 6.3
Maximum Current For Heatpump System		A	13.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 98	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -20 °C) *4 Heating (Circuit): 20 / 75 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 18.0 Heating: 18.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 31***	Heating: 31***	Heating: 31***
		Power Level dB	Cooling: 44***	Heating: 44***	Heating: 44***
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1293 (50-7/8)		
Net Weight		kg (lbs)	79 (174)		
Water Pipe Diameter	Room	mm (inch)	31 (1-1/4)		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		Brushless DC Motor (Sensorless vector control system)		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer (Range)		bar	0 ~ 4		
Protection Device		A	Earth Leakage Circuit Breaker (30 ~ 40)		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	3.00 / 85		
Tank Volume (Spec / Nett)		L	127 / 120		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat / Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	8.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		EN14511
	Volume	L	120
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.65
	Surface Area	m^2	1.3
	Total Length	m	18.5
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60ºC	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65ºC	µS/cm	< 1200
	Saturation Index (LSI) @ 20ºC		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in occordance with EU directive 2003/32/EC.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- *** The sound power level is measured with accordance to EN12102 under full load conditions. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- *3 When outdoor ambient is under -10°C, only the backup heater operate above 55°C.
- *4 Between outdoor ambient -10°C and -20°C, the water outlet temperature gradually decreases from 75°C to 55°C.

3.2 WH-ADC0509L3E51 WH-WDG07LE5

Item			Unit	Outdoor Unit		
Performance Test Condition				EN 14511		
				EN 14825		
Cooling Capacity			Condition (Ambient/Water)	A35W7		
			kW	7.00		
			BTU/h	23900		
Cooling EER			W/W	3.03		
Heating Capacity			Condition (Ambient/Water)	A7W35	A2W35	
			kW	7.00	6.85	
			BTU/h	23900	23400	
Heating COP			W/W	4.93	3.43	
Heating Erp	Low Temperature Application (W35)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	7.0	7.0	7.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-17 / -22
	SCOP / ns		(W/W) / %	6.31 / 249	4.96 / 195	4.25 / 167
	Annual Consumption		kWh	1483	2916	4060
	Class			A+++	A+++	A++
	Medium Temperature Application (W55)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	6.0	7.0	7.0
	Tbivalent / TOL		°C	2 / 2	-7 / -10	-15 / -22
	SCOP / ns		(W/W) / %	4.52 / 178	3.62 / 142	3.29 / 129
	Annual Consumption		kWh	1775	3991	5241
	Class			A+++	A++	A++
	DHW			Warmer	Average	Colder
	Application		Climate			
	COP / nwh		(W/W) / %	3.35 / 134	2.88 / 115	2.26 / 90
	AEC		kWh	384	447	569
Noise Level			Condition (Ambient/Water)	A35W7	A7W35	A2W35
			dB (A)	Cooling: -	Heating: -	Heating: -
			Power Level dB	Cooling: 63***	Heating: 60*** Heating: 53***	Heating: 60*** Heating: 53***
Air Flow			m^3/min (ft^3/min)	Cooling: 57.0 (2013) Heating: 51.0 (1801)		
Refrigeration Control Device				Expansion Valve		
Refrigeration Oil			cm^3	PZ68S (1100)		
Refrigerant			kg (oz)	R290, 0.96 (33.9) (Pre-charged) (-) (Maximum)		
F-GAS	GWP			3		
	CO^2eq (ton) (Precharged / Maximum)			0.003 / -		
Dimension	Height		mm (inch)	996 (39-7/32)		
	Width		mm (inch)	980 (38-37/64)		
	Depth		mm (inch)	430 (16-59/64)		
Net Weight			kg (lbs)	98 (216)		

Item		Unit	Outdoor Unit		
Pipe Diameter (Inner)		mm	25		
Standard Length		m (ft)	5.0 (16.4)		
Maximum Pipe Length		m (ft)	30.0 (98.4)		
I/D & O/D Height Difference		m (ft)	10 (32.8)		
Water Pipe Connector	Indoor	inch	1		
	Outdoor		1		
Compressor	Type		Hermetic Motor Compressor (Rotary)		
	Motor Type		Synchronous Electric Motor (4-poles)		
	Rated Output	kW	1.70		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	kW	1.562 (Heating) / 2.541 (Cooling)		
	Output Power	W	120		
	Fan Speed	rpm	Cooling: 540 Heating: 490		
Heat Exchanger	Fin material		Aluminium (Blue Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 46 × 19		
	Size (W × H × L)	mm	36.38 × 966 × 880.6 : 908.6		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		26		
	Size (W x H x L)	mm	57.8 × 524 × 117		
	Water Flow Rate	l/min (m^3/h)	Cooling: 20.1 (1.2) Heating: 20.1 (1.2)		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 2.31	Heating: 1.42	Heating: 2.00
Maximum Input Power For Heatpump System		kW	3.56		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 15.8 / 3.56k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	10.2		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 10.2	Heating: 6.3	Heating: 8.9
Maximum Current For Heatpump System		A	15.8		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 98	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -20 °C) *4 Heating (Circuit): 20 / 75 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 25.0 Heating: 25.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 31***	Heating: 31***	Heating: 31***
		Power Level dB	Cooling: 44***	Heating: 44***	Heating: 44***
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1293 (50-7/8)		
Net Weight		kg (lbs)	79 (174)		
Water Pipe Diameter	Room	mm (inch)	31 (1-1/4)		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		Brushless DC Motor (Sensorless vector control system)		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer (Range)		bar	0 ~ 4		
Protection Device		A	Earth Leakage Circuit Breaker (30 ~ 40)		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	3.00 / 85		
Tank Volume (Spec / Nett)		L	127 / 120		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat / Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	8.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		EN14511
	Volume	L	120
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.65
	Surface Area	m^2	1.3
	Total Length	m	18.5
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60ºC	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65ºC	µS/cm	< 1200
	Saturation Index (LSI) @ 20ºC		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in occordance with EU directive 2003/32/EC.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- *** The sound power level is measured with accordance to EN12102 under full load conditions. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- *3 When outdoor ambient is under -10°C, only the backup heater operate above 55°C.
- *4 Between outdoor ambient -10°C and -20°C, the water outlet temperature gradually decreases from 75°C to 55°C.

3.3 WH-ADC0509L3E51 WH-WDG09LE5

Item			Unit	Outdoor Unit		
Performance Test Condition				EN 14511		
				EN 14825		
Cooling Capacity			Condition (Ambient/Water)	A35W7		
			kW	8.20		
			BTU/h	28000		
Cooling EER			W/W	2.82		
Heating Capacity			Condition (Ambient/Water)	A7W35	A2W35	
			kW	9.00	7.00	
			BTU/h	30700	23900	
Heating COP			W/W	4.55	3.41	
Heating Erp	Low Temperature Application (W35)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	7.0	8.0	9.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns		(W/W) / %	6.44 / 255	4.84 / 190	4.31 / 170
	Annual Consumption		kWh	1453	3417	5142
	Class			A+++	A+++	A++
	Medium Temperature Application (W55)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	7.0	8.0	8.0
	Tbivalent / TOL		°C	2 / 2	-7 / -10	-15 / -22
	SCOP / ns		(W/W) / %	4.50 / 177	3.67 / 144	3.33 / 130
	Annual Consumption		kWh	2080	4499	5915
	Class			A+++	A++	A++
	DHW			Warmer	Average	Colder
	Application		Climate			
	COP / nwh		(W/W) / %	3.35 / 134	2.88 / 115	2.26 / 90
	AEC		kWh	384	447	569
Noise Level			Condition (Ambient/Water)	A35W7	A7W35	A2W35
			dB (A)	Cooling: -	Heating: -	Heating: -
			Power Level dB	Cooling: 63***	Heating: 63*** Heating: 54***	Heating: 63*** Heating: 54***
Air Flow			m^3/min (ft^3/min)	Cooling: 61.0 (2154) Heating: 74.0 (2613)		
Refrigeration Control Device				Expansion Valve		
Refrigeration Oil			cm^3	PZ68S (1100)		
Refrigerant			kg (oz)	R290, 1.00 (35.3) (Pre-charged) (-) (Maximum)		
F-GAS	GWP			3		
	CO^2eq (ton) (Precharged / Maximum)			0.003 / -		
Dimension	Height		mm (inch)	996 (39-7/32)		
	Width		mm (inch)	980 (38-37/64)		
	Depth		mm (inch)	430 (16-59/64)		
Net Weight			kg (lbs)	97 (214)		

Item		Unit	Outdoor Unit		
Pipe Diameter (Inner)		mm	25		
Standard Length		m (ft)	5.0 (16.4)		
Maximum Pipe Length		m (ft)	30.0 (98.4)		
I/D & O/D Height Difference		m (ft)	10 (32.8)		
Water Pipe Connector	Indoor	inch	1		
	Outdoor		1		
Compressor	Type		Hermetic Motor Compressor (Rotary)		
	Motor Type		Synchronous Electric Motor (4-poles)		
	Rated Output	kW	1.70		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	kW	2.178 (Heating) / 3.201 (Cooling)		
	Output Power	W	120		
	Fan Speed	rpm	Cooling: 570 Heating: 560		
Heat Exchanger	Fin material		Aluminium (Blue Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 46 × 19		
	Size (W × H × L)	mm	36.38 × 966 × 880.6 : 908.6		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		26		
	Size (W x H x L)	mm	57.8 × 524 × 117		
	Water Flow Rate	l/min (m^3/h)	Cooling: 23.5 (1.4) Heating: 25.8 (1.5)		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 2.91	Heating: 1.98	Heating: 2.05
Maximum Input Power For Heatpump System		kW	3.56		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 15.8 / 3.56k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	12.9		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 12.9	Heating: 8.8	Heating: 9.1
Maximum Current For Heatpump System		A	15.8		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 98	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -20 °C) *4 Heating (Circuit): 20 / 75 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 32.0 Heating: 36.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 31***	Heating: 31***	Heating: 31***
		Power Level dB	Cooling: 44***	Heating: 44***	Heating: 44***
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1293 (50-7/8)		
Net Weight		kg (lbs)	79 (174)		
Water Pipe Diameter	Room	mm (inch)	31 (1-1/4)		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		Brushless DC Motor (Sensorless vector control system)		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer (Range)		bar	0 ~ 4		
Protection Device		A	Earth Leakage Circuit Breaker (30 ~ 40)		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	3.00 / 85		
Tank Volume (Spec / Nett)		L	127 / 120		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat / Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	8.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		EN14511
	Volume	L	120
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.65
	Surface Area	m^2	1.3
	Total Length	m	18.5
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60ºC	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65ºC	µS/cm	< 1200
	Saturation Index (LSI) @ 20ºC		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in occordance with EU directive 2003/32/EC.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- *** The sound power level is measured with accordance to EN12102 under full load conditions. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- *3 When outdoor ambient is under -10°C, only the backup heater operate above 55°C.
- *4 Between outdoor ambient -10°C and -20°C, the water outlet temperature gradually decreases from 75°C to 55°C.

3.4 WH-ADC0509L3E5UK1 WH-WDG05LE5

Item			Unit	Outdoor Unit		
Performance Test Condition				EN 14511		
				EN 14825		
Cooling Capacity			Condition (Ambient/Water)	A35W7		
			kW	5.00		
			BTU/h	17100		
Cooling EER			W/W	3.23		
Heating Capacity			Condition (Ambient/Water)	A7W35	A2W35	
			kW	5.00	5.00	
			BTU/h	17100	17100	
Heating COP			W/W	5.05	3.52	
Heating Erp	Low Temperature Application (W35)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	5.0	5.0	6.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns		(W/W) / %	6.00 / 237	5.06 / 200	4.25 / 167
	Annual Consumption		kWh	1113	2040	3483
	Class			A+++	A+++	A++
	Medium Temperature Application (W55)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	5.0	5.0	6.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns		(W/W) / %	4.27 / 168	3.63 / 142	3.28 / 128
	Annual Consumption		kWh	1565	2849	4516
	Class			A+++	A++	A++
	DHW			Warmer	Average	Colder
	Application		Climate			
	COP / nwh		(W/W) / %	3.35 / 134	2.88 / 115	2.26 / 90
	AEC		kWh	384	447	569
Noise Level			Condition (Ambient/Water)	A35W7	A7W35	A2W35
			dB (A)	Cooling: -	Heating: -	Heating: -
			Power Level dB	Cooling: 61***	Heating: 57*** Heating: 52***	Heating: 57*** Heating: 52***
Air Flow			m^3/min (ft^3/min)	Cooling: 55.0 (1942) Heating: 45.0 (1589)		
Refrigeration Control Device				Expansion Valve		
Refrigeration Oil			cm^3	PZ68S (1100)		
Refrigerant			kg (oz)	R290, 0.96 (33.9) (Pre-charged) (-) (Maximum)		
F-GAS	GWP			3		
	CO^2eq (ton) (Precharged / Maximum)			0.003 / -		
Dimension	Height		mm (inch)	996 (39-7/32)		
	Width		mm (inch)	980 (38-37/64)		
	Depth		mm (inch)	430 (16-59/64)		
Net Weight			kg (lbs)	98 (216)		

Item		Unit	Outdoor Unit		
Pipe Diameter (Inner)		mm	20		
Standard Length		m (ft)	5.0 (16.4)		
Maximum Pipe Length		m (ft)	30.0 (98.4)		
I/D & O/D Height Difference		m (ft)	10 (32.8)		
Water Pipe Connector	Indoor	inch	1		
	Outdoor		1		
Compressor	Type		Hermetic Motor Compressor (Rotary)		
	Motor Type		Synchronous Electric Motor (4-poles)		
	Rated Output	kW	1.70		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	kW	1.089 (Heating) / 1.705 (Cooling)		
	Output Power	W	120		
	Fan Speed	rpm	Cooling: 520 Heating: 440		
Heat Exchanger	Fin material		Aluminium (Blue Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 46 × 19		
	Size (W × H × L)	mm	36.38 × 966 × 880.6 : 908.6		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		26		
	Size (W x H x L)	mm	57.8 × 524 × 117		
	Water Flow Rate	l/min (m^3/h)	Cooling: 14.3 (0.9) Heating: 14.3 (0.9)		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 1.55	Heating: 0.99	Heating: 1.42
Maximum Input Power For Heatpump System		kW	2.93		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 2.93k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	6.9		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 6.9	Heating: 4.4	Heating: 6.3
Maximum Current For Heatpump System		A	13.0		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 98	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -20 °C) *4 Heating (Circuit): 20 / 75 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 18.0 Heating: 18.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 31***	Heating: 31***	Heating: 31***
		Power Level dB	Cooling: 44***	Heating: 44***	Heating: 44***
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1293 (50-7/8)		
Net Weight		kg (lbs)	79 (174)		
Water Pipe Diameter	Room	mm (inch)	31 (1-1/4)		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		Brushless DC Motor (Sensorless vector control system)		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer (Range)		bar	0 ~ 4		
Protection Device		A	Earth Leakage Circuit Breaker (30 ~ 40)		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	3.00 / 85		
Tank Volume (Spec / Nett)		L	127 / 120		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat / Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	10.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		EN14511
	Volume	L	120
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.65
	Surface Area	m^2	1.3
	Total Length	m	18.5
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60ºC	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65ºC	µS/cm	< 1200
	Saturation Index (LSI) @ 20ºC		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in occordance with EU directive 2003/32/EC.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- *** The sound power level is measured with accordance to EN12102 under full load conditions. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- *3 When outdoor ambient is under -10°C, only the backup heater operate above 55°C.
- *4 Between outdoor ambient -10°C and -20°C, the water outlet temperature gradually decreases from 75°C to 55°C.

3.5 WH-ADC0509L3E5UK1 WH-WDG07LE5

Item			Unit	Outdoor Unit		
Performance Test Condition				EN 14511		
				EN 14825		
Cooling Capacity			Condition (Ambient/Water)	A35W7		
			kW	7.00		
			BTU/h	23900		
Cooling EER			W/W	3.03		
Heating Capacity			Condition (Ambient/Water)	A7W35	A2W35	
			kW	7.00	6.85	
			BTU/h	23900	23400	
Heating COP			W/W	4.93	3.43	
Heating Erp	Low Temperature Application (W35)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	7.0	7.0	7.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-17 / -22
	SCOP / ns		(W/W) / %	6.31 / 249	4.96 / 195	4.25 / 167
	Annual Consumption		kWh	1483	2916	4060
	Class			A+++	A+++	A++
	Medium Temperature Application (W55)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	6.0	7.0	7.0
	Tbivalent / TOL		°C	2 / 2	-7 / -10	-15 / -22
	SCOP / ns		(W/W) / %	4.52 / 178	3.62 / 142	3.29 / 129
	Annual Consumption		kWh	1775	3991	5241
	Class			A+++	A++	A++
	DHW			Warmer	Average	Colder
	Application		Climate			
	COP / nwh		(W/W) / %	3.35 / 134	2.88 / 115	2.26 / 90
	AEC		kWh	384	447	569
Noise Level			Condition (Ambient/Water)	A35W7	A7W35	A2W35
			dB (A)	Cooling: -	Heating: -	Heating: -
			Power Level dB	Cooling: 63***	Heating: 60*** Heating: 53***	Heating: 60*** Heating: 53***
Air Flow			m^3/min (ft^3/min)	Cooling: 57.0 (2013) Heating: 51.0 (1801)		
Refrigeration Control Device				Expansion Valve		
Refrigeration Oil			cm^3	PZ68S (1100)		
Refrigerant			kg (oz)	R290, 0.96 (33.9) (Pre-charged) (-) (Maximum)		
F-GAS	GWP			3		
	CO^2eq (ton) (Precharged / Maximum)			0.003 / -		
Dimension	Height		mm (inch)	996 (39-7/32)		
	Width		mm (inch)	980 (38-37/64)		
	Depth		mm (inch)	430 (16-59/64)		
Net Weight			kg (lbs)	98 (216)		

Item		Unit	Outdoor Unit		
Pipe Diameter (Inner)		mm	25		
Standard Length		m (ft)	5.0 (16.4)		
Maximum Pipe Length		m (ft)	30.0 (98.4)		
I/D & O/D Height Difference		m (ft)	10 (32.8)		
Water Pipe Connector	Indoor	inch	1		
	Outdoor		1		
Compressor	Type		Hermetic Motor Compressor (Rotary)		
	Motor Type		Synchronous Electric Motor (4-poles)		
	Rated Output	kW	1.70		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	kW	1.562 (Heating) / 2.541 (Cooling)		
	Output Power	W	120		
	Fan Speed	rpm	Cooling: 540 Heating: 490		
Heat Exchanger	Fin material		Aluminium (Blue Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 46 × 19		
	Size (W × H × L)	mm	36.38 × 966 × 880.6 : 908.6		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		26		
	Size (W x H x L)	mm	57.8 × 524 × 117		
	Water Flow Rate	l/min (m^3/h)	Cooling: 20.1 (1.2) Heating: 20.1 (1.2)		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 2.31	Heating: 1.42	Heating: 2.00
Maximum Input Power For Heatpump System		kW	3.56		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 15.8 / 3.56k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	10.2		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 10.2	Heating: 6.3	Heating: 8.9
Maximum Current For Heatpump System		A	15.8		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 98	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -20 °C) *4 Heating (Circuit): 20 / 75 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 25.0 Heating: 25.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 31***	Heating: 31***	Heating: 31***
		Power Level dB	Cooling: 44***	Heating: 44***	Heating: 44***
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1293 (50-7/8)		
Net Weight		kg (lbs)	79 (174)		
Water Pipe Diameter	Room	mm (inch)	31 (1-1/4)		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		Brushless DC Motor (Sensorless vector control system)		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer (Range)		bar	0 ~ 4		
Protection Device		A	Earth Leakage Circuit Breaker (30 ~ 40)		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	3.00 / 85		
Tank Volume (Spec / Nett)		L	127 / 120		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat / Cool	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	10.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		EN14511
	Volume	L	120
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.65
	Surface Area	m^2	1.3
	Total Length	m	18.5
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60ºC	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65ºC	µS/cm	< 1200
	Saturation Index (LSI) @ 20ºC		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in occordance with EU directive 2003/32/EC.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- *** The sound power level is measured with accordance to EN12102 under full load conditions. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- *3 When outdoor ambient is under -10°C, only the backup heater operate above 55°C.
- *4 Between outdoor ambient -10°C and -20°C, the water outlet temperature gradually decreases from 75°C to 55°C.

3.6 WH-ADC0509L3E5UK1 WH-WDG09LE5

Item			Unit	Outdoor Unit		
Performance Test Condition				EN 14511		
				EN 14825		
Cooling Capacity			Condition (Ambient/Water)	A35W7		
			kW	8.20		
			BTU/h	28000		
Cooling EER			W/W	2.82		
Heating Capacity			Condition (Ambient/Water)	A7W35	A2W35	
			kW	9.00	7.00	
			BTU/h	30700	23900	
Heating COP			W/W	4.55	3.41	
Heating Erp	Low Temperature Application (W35)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	7.0	8.0	9.0
	Tbivalent / TOL		°C	2 / 2	-10 / -10	-15 / -22
	SCOP / ns		(W/W) / %	6.44 / 255	4.84 / 190	4.31 / 170
	Annual Consumption		kWh	1453	3417	5142
	Class			A+++	A+++	A++
	Medium Temperature Application (W55)			Warmer	Average	Colder
	Application		Climate			
	Pdesign		kW	7.0	8.0	8.0
	Tbivalent / TOL		°C	2 / 2	-7 / -10	-15 / -22
	SCOP / ns		(W/W) / %	4.50 / 177	3.67 / 144	3.33 / 130
	Annual Consumption		kWh	2080	4499	5915
	Class			A+++	A++	A++
	DHW			Warmer	Average	Colder
	Application		Climate			
	COP / nwh		(W/W) / %	3.35 / 134	2.88 / 115	2.26 / 90
	AEC		kWh	384	447	569
Noise Level			Condition (Ambient/Water)	A35W7	A7W35	A2W35
			dB (A)	Cooling: -	Heating: -	Heating: -
			Power Level dB	Cooling: 63***	Heating: 63*** Heating: 54***	Heating: 63*** Heating: 54***
Air Flow			m^3/min (ft^3/min)	Cooling: 61.0 (2154) Heating: 74.0 (2613)		
Refrigeration Control Device				Expansion Valve		
Refrigeration Oil			cm^3	PZ68S (1100)		
Refrigerant			kg (oz)	R290, 1.00 (35.3) (Pre-charged) (-) (Maximum)		
F-GAS	GWP			3		
	CO^2eq (ton) (Precharged / Maximum)			0.003 / -		
Dimension	Height		mm (inch)	996 (39-7/32)		
	Width		mm (inch)	980 (38-37/64)		
	Depth		mm (inch)	430 (16-59/64)		
Net Weight			kg (lbs)	97 (214)		

Item		Unit	Outdoor Unit		
Pipe Diameter (Inner)		mm	25		
Standard Length		m (ft)	5.0 (16.4)		
Maximum Pipe Length		m (ft)	30.0 (98.4)		
I/D & O/D Height Difference		m (ft)	10 (32.8)		
Water Pipe Connector	Indoor	inch	1		
	Outdoor		1		
Compressor	Type		Hermetic Motor Compressor (Rotary)		
	Motor Type		Synchronous Electric Motor (4-poles)		
	Rated Output	kW	1.70		
Fan	Type		Propeller Fan		
	Material		PP		
	Motor Type		DC (8-poles)		
	Input Power	kW	2.178 (Heating) / 3.201 (Cooling)		
	Output Power	W	120		
	Fan Speed	rpm	Cooling: 570 Heating: 560		
Heat Exchanger	Fin material		Aluminium (Blue Coat)		
	Fin Type		Corrugated Fin		
	Row × Stage × FPI		2 × 46 × 19		
	Size (W × H × L)	mm	36.38 × 966 × 880.6 : 908.6		
Hot Water Coil	Type		Brazed Plate		
	No. of Plates		26		
	Size (W x H x L)	mm	57.8 × 524 × 117		
	Water Flow Rate	l/min (m^3/h)	Cooling: 23.5 (1.4) Heating: 25.8 (1.5)		
Power Source (Phase, Voltage, Cycle)		Ø	Single		
		V	230		
		Hz	50		
Input Power		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		kW	Cooling: 2.91	Heating: 1.98	Heating: 2.05
Maximum Input Power For Heatpump System		kW	3.56		
Power Supply 1 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 15.8 / 3.56k		
Power Supply 2 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			1Ø / 13.0 / 3.00k		
Power Supply 3 : Phase (Ø) / Max. Current (A) / Max. Input Power (W)			- / - / -		
Starting Current		A	12.9		
Running Current		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		A	Cooling: 12.9	Heating: 8.8	Heating: 9.1
Maximum Current For Heatpump System		A	15.8		
Power Factor Power factor means total figure of compressor and outdoor fan motor.		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		%	Cooling: 98	Heating: 98	Heating: 98
Power Cord	Number of core		-		
	Length	m (ft)	-		
Thermostat			Electronic Control		
Protection Device			Electronic Control		
Pressure Relief Valve Water Circuit		kPa	Open: 300, Close: 210 and below		

Item		Unit	Indoor Unit		
Performance Test Condition			EN 14511		
			EN 14825		
Operation Range	Outdoor Ambient	°C (min. / max.)	Cooling: 10 / 43 Heating (Tank): -25 / 35 Heating (Circuit): -25 / 35		
	Water Outlet	°C (min. / max.)	Cooling: 5 / 20 Heating (Tank): - / 65*3, Heating (Circuit): 20 / 55 (Below Ambient -20 °C) *4 Heating (Circuit): 20 / 75 (Above Ambient -10 °C) *4		
Internal Pressure Differential		kPa	Cooling: 32.0 Heating: 36.0		
Noise Level		Condition (Ambient/Water)	A35W7	A7W35	A2W35
		dB (A)	Cooling: 31***	Heating: 31***	Heating: 31***
		Power Level dB	Cooling: 44***	Heating: 44***	Heating: 44***
Dimension	Depth	mm (inch)	602 (23-45/64)		
	Width	mm (inch)	599 (23-37/64)		
	Height	mm (inch)	1293 (50-7/8)		
Net Weight		kg (lbs)	79 (174)		
Water Pipe Diameter	Room	mm (inch)	31 (1-1/4)		
	Shower	mm (inch)	19 (3/4)		
Water Drain Hose Inner Diameter		mm (inch)	12.00 (17/36)		
Pump	Motor Type		Brushless DC Motor (Sensorless vector control system)		
	No. of Speed		7 (Software Selection)		
	Input Power	W	145		
Flow Sensor	Type		Vortex (Piezoelectric sensor)		
	Measuring range	l/min	5 ~ 60		
Pressure Transducer (Range)		bar	0 ~ 4		
Protection Device		A	Earth Leakage Circuit Breaker (30 ~ 40)		
Expansion Vessel	Volume	l	10		
	MWP	bar	3		
Capacity of Integrated Electric Heater / OLP TEMP		kW / °C	3.00 / 85		
Tank Volume (Spec / Nett)		L	127 / 120		
Max. Tank Water Set Temperature		°C	65		
Tank Coil Surface		m^2	1.8		
Maximum Working Pressure	Heat / Cool *10	Bar	3.0		
	Tank Circuit	Bar	10.0		
Operating Pressure	Tank Unit	Bar	3.5		
	Expansion Relief Valve	Bar	10.0		
Expansion Vessel Pre-charge Pressure (DHW Circuit)		Bar	3.5		
Pressure Reducing Valve Set Pressure (DHW Circuit)		Bar	3.5		

Item		Unit	Indoor Unit
Pressure Vessel	Material		EN14511
	Volume	L	120
	Design Pressure	Bar	10
Heat Exchanger	Material		EN-1.4521
	Diameter	mm	22
	Thickness	mm	0.65
	Surface Area	m^2	1.3
	Total Length	m	18.5
DHW Tank	Total Corrosion ion (Chloride + Sulphate + Nitric)	mg/L	< 150
	Conductivity @ Water Tank Water Temperature < 60ºC	µS/cm	< 1250
	Conductivity @ Water Tank Water Temperature < 65ºC	µS/cm	< 1200
	Saturation Index (LSI) @ 20ºC		> -4.0 / < 0.4
	PH		6.5 - 8.5

Note:

- In case it is necessary to indicate the air flow volume in (l/s), the value in (m^3/min.) shall be multiplied by 16.7 and rounded down the decimal point.
- If the EUROVENT Certified models can be operated under the “extra-low” temperature condition, -7°C DB and -8°C WB temperature with rated voltage 230V shall be used.
- Capacity is measured at outdoor temperature 7°C DB and 6°C WB with controlled water inlet 30°C and water outlet 35°C (EN 14511-2)
- Flowrate indicated are based on nominal capacity adjustment of leaving water temperature (LWT) 35°C and ΔT=5°C.
- EER and COP classification is at 230V only in occordance with EU directive 2003/32/EC.
- *** The sound pressure and sound power level is measured with distance 1.0m from the unit and height at 1.5m. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- **** The sound power level is measured with accordance to EN12102 under conditions of the EN14825.
- *** The sound power level is measured with accordance to EN12102 under full load conditions. (Test carry out for cooling at ambient 35°C DB and Water Out 7°C, heating at ambient 7°C DB / 6°C WB and water out 55°C)
- *3 When outdoor ambient is under -10°C, only the backup heater operate above 55°C.
- *4 Between outdoor ambient -10°C and -20°C, the water outlet temperature gradually decreases from 75°C to 55°C.

4. Features

- **Inverter Technology**
  - Energy saving

- **High Efficiency**

- **Long Installation Piping**
  - Long piping up to 30 meter

- **A-class energy efficiency pump**
  - Water pump speed can be set by selection at control panel

- **Improved deice cycle**

- **Protection Feature**
  - Random auto restart after power failure for safety restart operation
  - Gas leakage protection
  - Prevent compressor reverse cycle
  - Inner protector to protect compressor

- **Serviceability Feature**
  - Breakdown Self Diagnosis function
  - System Status Check Buttons for servicing purpose
  - System Pumpdown Button for servicing purpose
  - Front maintenance design for outdoor unit

5. Location of Controls and Components

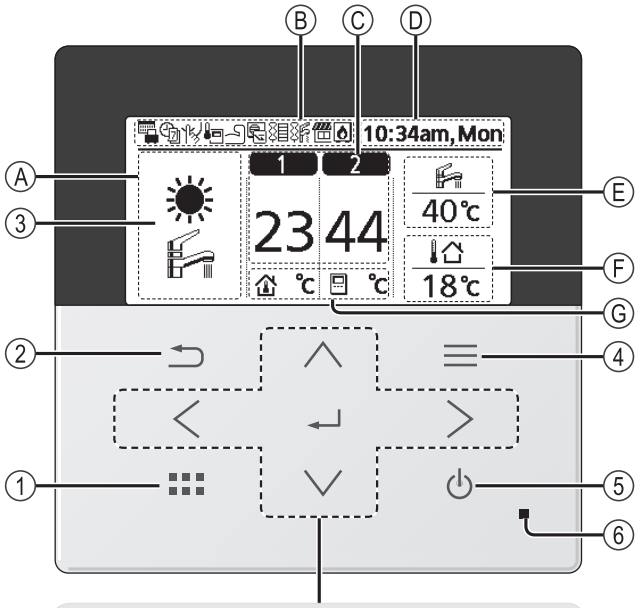
5.1 Indoor Unit

5.1.1 Remote Controller Buttons and Display

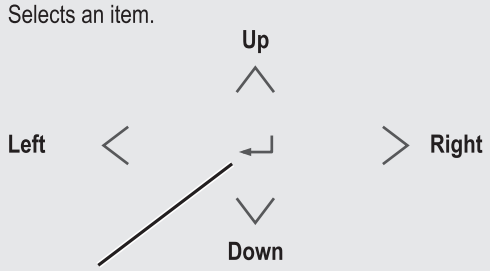
The LCD display as shown in this manual are for instructional purpose only, and may differ from the actual unit.

Buttons / Indicator	
①	**Quick Menu button**
②	**Back button** Returns to the previous screen
③	**LCD Display** (Actual - Dark background with white icons)
④	**Main Menu button** For function setup
⑤	**ON/OFF button** Starts/Stops operation
⑥	**Operation indicator** Illuminates during operation, blinks during alarm.

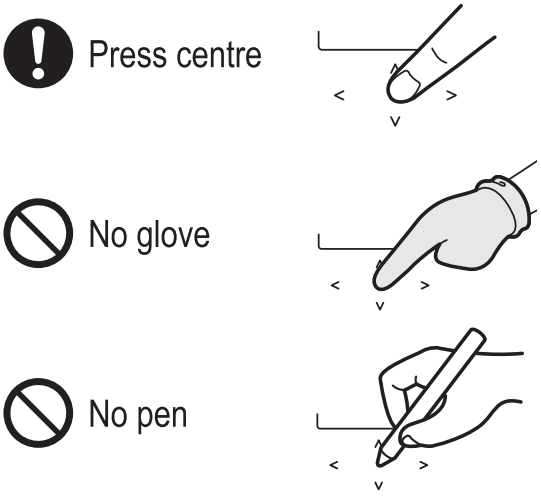
When the backlight is off, press any button to turn it on.
(Do not press button ⑤)
The time until the backlight turns off can be changed in the Menu (Personal setup)



Cross key buttons
Selects an item.

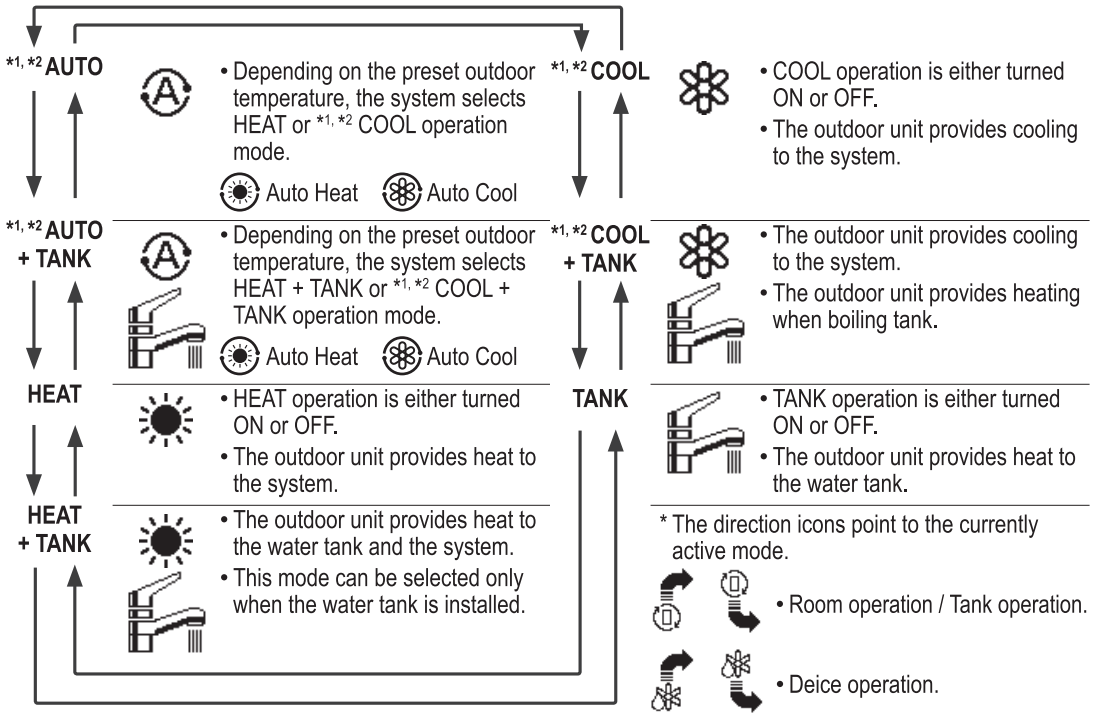


Enter button
Fixes the selected content.



Display

Ⓐ **Mode selection**



Ⓑ **Operation icons**

The status of operation is displayed.
Icon will not display (under operation OFF screen) whenever operation is OFF except weekly timer.

- Holiday operation status
- Weekly Timer operation status
- Quiet operation status
- Zone:Room Thermostat →Internal sensor status
- Powerful operation status
- Demand Control or SG ready or SHP status
- Room Heater status
- Tank Heater status
- Solar status
- Bivalent status (Boiler)

Ⓒ **Temperature of each zone**

Ⓓ **Time and day**

Ⓔ **Water Tank temperature**

Ⓕ **Outdoor temperature**

Ⓖ **Sensor type/Set temperature type icons**

- Water Temperature →Compensation curve
- Water Temperature →Direct
- Pool only
- Room Thermostat →External
- Room Thermostat →Internal
- Room Thermistor

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

5.1.2 Initialization

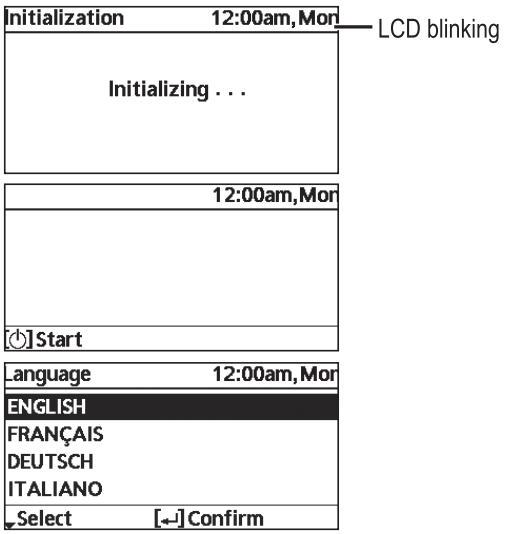
Before starting to install the various menu settings, please initiate the Remote Controller by selecting the language of operation and installing the date and time correctly.
When power is turned on for the first time, it becomes the setting screen automatically. It can also be set from personal setting of the menu.

Selecting the language

Wait while the display is initializing.
When initializing screen ends, it turns to normal screen.
When any button is pressed, language setting screen appears.

① Scroll with ∨ and ∧ to select the language.

② Press ↵ to confirm the selection.



Setting the clock

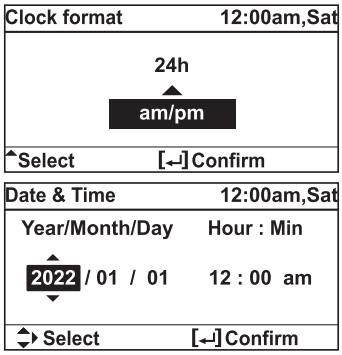
① Select with ∨ or ∧ how to display the time, either 24h or am/pm format (for example, 15:00 or 3:00 pm).

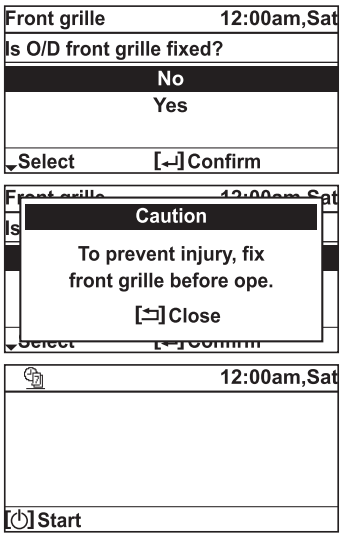
② Press ↵ to confirm the selection.

③ Use ∨ and ∧ to select year, month, day, hour and minutes. (Select and move with > and press ↵ to confirm.)

④ Once the time is set, time and day will appear on the display even if the Remote Controller is turned OFF.

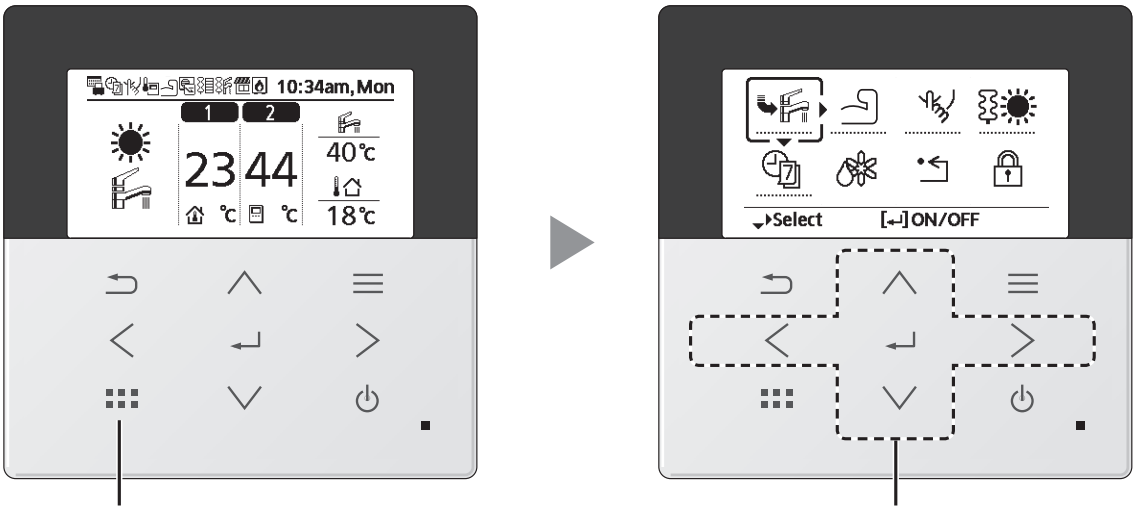
⑤ Final precaution step to check and confirm whether outdoor front grille is fixed before operating the unit for safety purpose.
Select Yes if outdoor front grille is already fixed. Then it will proceed to main screen.
Select No if outdoor front grille is not yet fixed. A caution message will pop up to remind on the installation.





5.1.3 Quick Menu

After the initial settings have been completed, you can select a quick menu from the following options and edit the setting.



① Press ▪▪▪ to display the quick menu.

② Use ∧ ∨ < > to select menu.

③ Press ↵ to turn on/off the select menu.

Quick Menu

Force DHW	Powerful	Quiet	Force Heater
Weekly Timer	Force Defrost	Error Reset	R/C Lock

▾Select [↵] ON/OFF	Select each setting and confirm the setting according to the instructions displayed at the bottom of the screen. (The icons refer to each selection key.)

To return to the Main Screen,

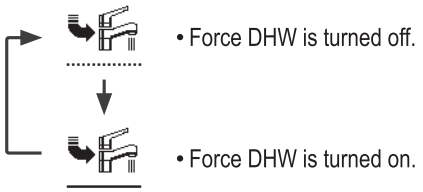
Press ▪▪▪ or ⮌ .

5.1.4 How to Use the Quick Menu

Force DHW

Select this icon to turn the Tank DHW on or off.

Press ↵ to confirm your selection.



• Force DHW is turned off.

• Force DHW is turned on.

Note:

• Force DHW is disabled when Force Heater is turned on.

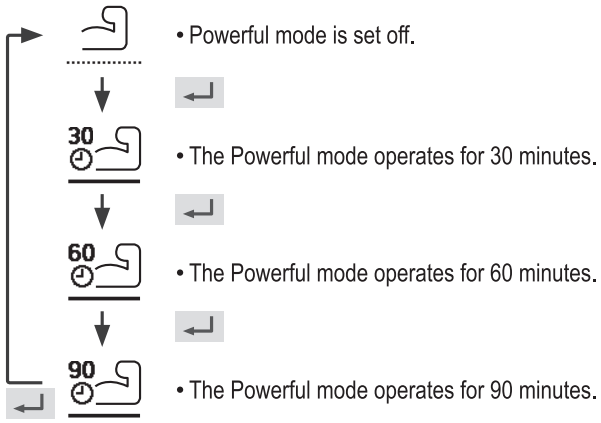
• When Force DHW is turned off, operation & mode should change back to the previous memorized status.

Powerful

Select this icon to operate the heating/cooling system powerfully.

Press ↵ to confirm your selection.

(The powerful operation starts approximately 1 minute after ↵ is pressed.)



• Powerful mode is set off.

• The Powerful mode operates for 30 minutes.

• The Powerful mode operates for 60 minutes.

• The Powerful mode operates for 90 minutes.

Note:

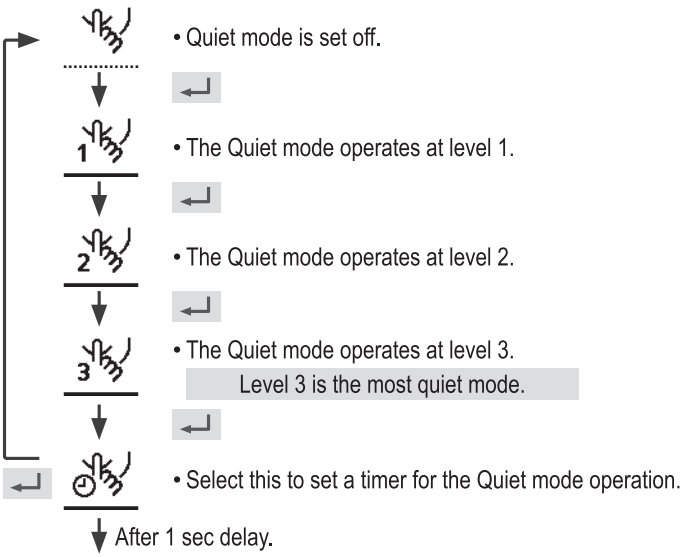
• Powerful is disabled when operation is turned OFF.

Quiet

Select this icon to operate quietly.

Press ↵ to confirm your selection.

(The quiet operation starts approximately 1 minute after ↵ is pressed.)



- Quiet mode is set off.
- The Quiet mode operates at level 1.
- The Quiet mode operates at level 2.
- The Quiet mode operates at level 3.
  Level 3 is the most quiet mode.
- Select this to set a timer for the Quiet mode operation.

After 1 sec delay.



Select "Yes".

- Select "Yes" using < > buttons.

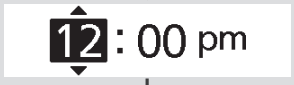


Select pattern "1" ~ "6".



Select "Edit".

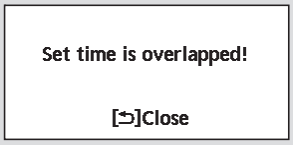
- If you select "Delete", the timer setting of the selected pattern will be deleted.



Set the hour and minutes.



Select the level of Quiet.



Note:

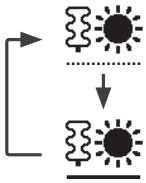
- If the time overlaps with another pattern, "Set time is overlapped!" will appear on the screen.

Force Heater

Select to force the Heater on.

Press ↵ to confirm your selection.

(The Force Heater mode starts approximately 1 minute after ↵ is pressed.)

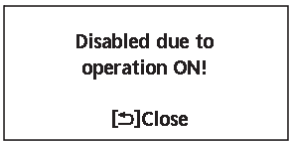


• Force Heater is turned off.

• Force Heater is turned on.

Note:

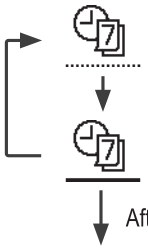
• Force Heater is disabled whenever operation is already on and “Disabled due to operation ON!” will be displayed.



Weekly Timer

Select this icon to delete (cancel) or change the pre-set Weekly Timer.

Press ↵ to confirm your selection.



• Timer is not ON.

• Timer is set.

After 1 sec delay.



Select "Yes".

• If you select "No", the screen will return to the Main Screen.



• Timer setup: Select Timer setup to edit the Weekly Timer.

• Timer copy: Select to copy a timer setting.

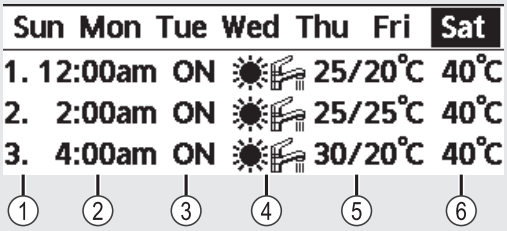


[Example of a Timer setup]

Select the day(s) which you wish to edit using ∧ ∨ buttons.



If all 6 patterns are not preset, this screen will be displayed.



① **Select pattern "1" ~ "6".**

② **Set the hour and minutes of the Timer.**

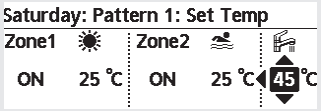
③ **Select ON/OFF of the Timer.**

④ **Select the operation mode.**

Ⓐ / Ⓐ / ☀ / ☀ / / ❄ / ❄

• Select mode using ∧ ∨ buttons.

⑤ **Set the temperature for both Zone 1 and 2 (if your system has the 2-Zone setting).**



⑥ **Set the Tank temperature.**

Note:

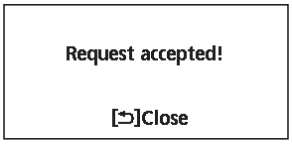
• Timer is disabled when Force Heater is turned on or Heat-Cool SW is enabled.

• If you have preset the Weekly Timer on 2 zones, you must repeat the same procedure with Zone 2.

Force Defrost

Select to defrost the frozen pipes.

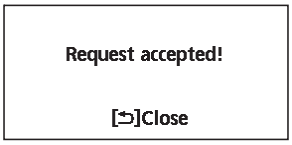
Press ↵ to confirm your selection.
(When the mode is accepted, below screen will be displayed.)



Error Reset

Select to restore the previous settings when error has occurred.

Press ↵ to confirm your selection.
(When the mode has been accepted, below screen will be displayed.)

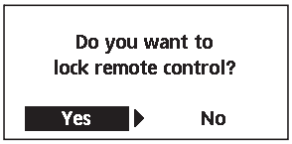


- Make sure all units are turned off before selecting this mode which restores the whole system to the previous settings.

R/C Lock

Select to lock the Remote Controller.

Press ↵ to confirm your selection.
(When the mode has been accepted, below screen will be displayed.)



Select "Yes".
(The Main Screen will be locked.)
- If "No" is selected, the screen will return to the Main Screen.

To unlock the Remote Controller

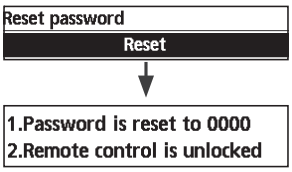
Press any key.
(When the mode has been accepted, below screen will be displayed.)



Enter any 4 digits of number (if the number is correct, the screen will be unlocked).

To reset forgotten password (under operation OFF screen)

Press ⮌ , ↵ and > continuously for 5 seconds.
(When the mode has been accepted, below screen will be displayed.)



Select "Reset".

(The screen will be off after 3 seconds.)

5.1.5 Menus (For User)

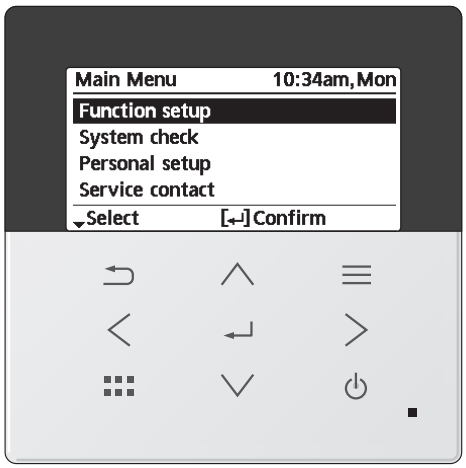
Select menus and determine settings according to the system available in the household. All initial settings must be done by an authorised dealer or a specialist. It is recommended that all alterations of the initial settings are also done by an authorised dealer or a specialist.

- After initial installation, you may manually adjust the settings.
- The initial setting remains active until the user changes it.
- The Remote Controller can be used for multiple installations.
- Ensure the operation indicator is OFF before setting.
- The system may not work properly if set wrongly. Please consult an authorised dealer.

To display <Main Menu>:

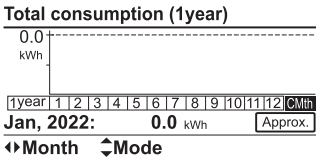
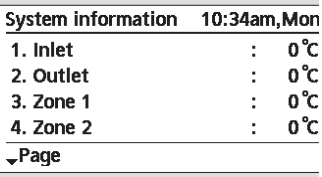
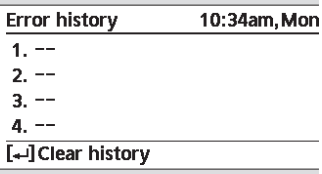
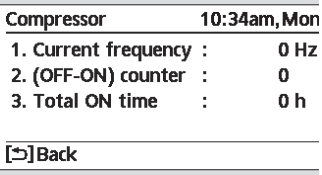
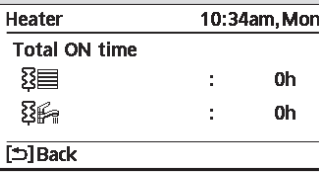
To select menu:

To confirm the selected content:

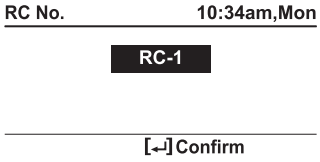
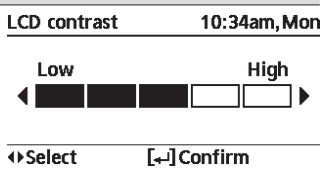
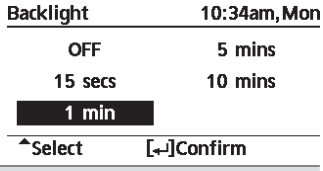
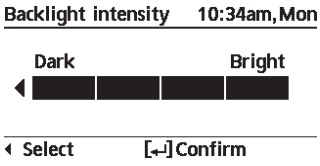
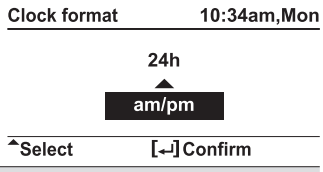
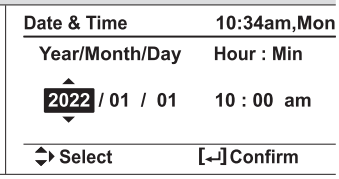


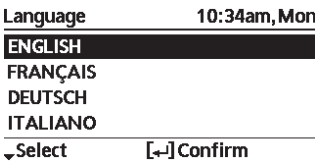
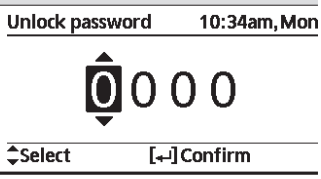
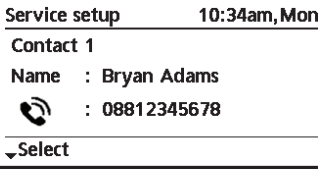
Menu	Default Setting	Setting Options / Display
1 Function setup		
1.1 > Weekly timer		
Once the weekly timer is set up, User can edit from Quick Menu. To set up to 6 patterns of operation on a daily basis. • Disabled if Heat-Cool SW is select "Yes" or if Force Heater is on.	**Timer setup** Select day of the week and set the patterns needed (Time / Operation ON/OFF / Mode)	Weekly timer 10:34am,Mon Sun Mon Tue Wed Thu Fri Sat 1. 8:00am ON 40℃ 2. 12:00pm ON 24/28℃ 40℃ 3. 1:00pm ON 12/10℃ Day Pattern [↵]Edit
	Timer copy Select day of the week	
1.2 > Holiday timer		
To save energy, a holiday period may be set to either turn OFF the system or lower the temperature during the period.	OFF	ON OFF
	> ON	
	Holiday start and end. Date and time	Holiday: End 10:34am,Mon Year/Month/Day Hour : Min 2022 / 01 / 01 10 : 00 am Select [↵]Confirm
	OFF or lowered temperature	
• Weekly timer setting may be temporarily disabled during Holiday timer setting but it will be restored once the Holiday timer is completed.		
1.3 > Quiet timer		
To operate quietly during the preset period. 6 patterns may be set. Level 0 means the mode is off.	Time to start Quiet : Date and time	Quiet 10:34am,Mon Pattern Time Level 1 8:00am 0 2 5:00pm 1 3 11:00pm 3 Select [↵]Edit
	Level of quietness: 0 ~ 3	

Menu	Default Setting	Setting Options / Display
1.4 > Quiet priority		
• To select priority during Quiet mode between Sound and Capacity. • If Sound priority is selected, unit will operate in quiet condition only. • If Capacity priority is selected, unit will operate in quiet condition but it will prioritize on providing required capacity at the same time.	Sound	Sound Capacity
1.5 > Room heater		
To set the room heater ON or OFF.	OFF	ON OFF
1.6 > Tank heater		
To set the tank heater ON or OFF.	OFF	ON OFF
1.7 > Sterilization		
To set the auto sterilization ON or OFF.	ON	ON OFF
• Do not use the system during sterilization in order to prevent scalding with hot water, or overheating of shower. • Ask an authorised dealer to determine the level of sterilization function field settings according to the local laws and regulations.		
1.8 > DHW mode (Domestic Hot Water)		
To set the DHW mode to Standard or Smart. • Standard mode have faster DHW Tank heat up time. Meanwhile Smart mode take longer time to heat up DHW time with lower energy consumption.	Standard	Standard Smart
To set the tank sensor to Top or Center. • Selection of the tank sensor to top slow down the start of boiling up the tank and reduce power consumption. Please change this selection to “Center” when the hot water becomes insufficient.	Top	Top Center

Menu	Default Setting	Setting Options / Display
2 System check		
2.1 > Energy monitor		
Present or historical chart of energy consumption, generation or COP.	**Present** Select and retrieve **Historical chart** Select and retrieve	
• COP= Coefficient of Performance. • For historical chart, the period is selected from 1 day/1 week/1year. • Energy consumption (kWh) of heating, *1, *2 cooling, tank and total may be retrieved. • The total power consumption is an estimated value based on AC 230 V and may differ from value measured by precise equipment.		
2.2 > System information		
Shows all system information in each area.	**Actual system information of 11 items: Inlet / Outlet / Zone 1 / Zone 2 / Tank / Buffer tank / Solar / Pool / COMP frequency / Pump flowrate / Water pressure** Select and retrieve	
2.3 > Error history		
• Refer to Troubleshooting for error codes. • The most recent error code is displayed at the top.	Select and retrieve	
2.4 > Compressor		
Shows the compressor performance.	Select and retrieve	
2.5 > Heater		
Total hours of ON time for Room heater/Tank heater.	Select and retrieve	

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
3 Personal setup		
3.1 > Remote control No.		
• To display remote control number of a particular remote controller so that installer and end user are well informed. • Main remote controller is displayed as RC-1. Second remote controller is displayed as RC-2.	Select and retrieve	
3.2 > Touch sound		
Turns the operation sound ON/ OFF.	ON	
3.3 > LCD contrast		
Sets the screen contrast.	3	
3.4 > Backlight		
Sets the duration of screen backlight.	1 min	
3.5 > Backlight intensity		
Sets screen backlight brightness.	4	
3.6 > Clock format		
Sets the type of clock display.	am/pm	
3.7 > Date & Time		
Sets the present date and time.	Year / Month / Day / Hour / Min	

Menu	Default Setting	Setting Options / Display
3.8 > Language		
Sets the display language for the top screen.	ENGLISH / FRANÇAIS / DEUTSCH / ITALIANO / ESPAÑOL / DANISH / SWEDISH / NORWEGIAN / POLISH / CZECH / NEDERLANDS / TÜRKÇE / SUOMI / MAGYAR / SLOVENŠČINA / HRVATSKI / LIETUVIŲ / PORTUGUÊS / БЪЛГАРСКИ / EESTI / LATVIEŠU / ROMÂNĂ / SHQIP / SLOVENČINA / МАКЕДОНСКИ / УКРАЇНСЬКА / ΕΛΛΗΝΙΚΑ	
3.9 > Unlock password		
4 digit password for all the settings.	0000	
4 Service contact		
4.1 > Contact 1 / Contact 2		
Preset contact number for installer.	Select and retrieve	

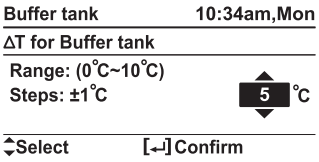
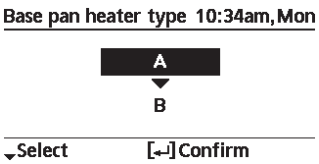
5.1.6 Menus (For Installer)

Menu	Default Setting	Setting Options / Display
5 Installer setup > System setup		
5.1 > Optional PCB connectivity		
To connect to the external PCB required for servicing.	No	Yes ▲ No

• If the external PCB is connected (optional), the system will have following additional functions:

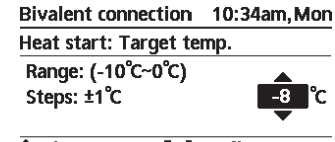
① Control over 2 zones (including the swimming pool and the function to heat water in it).
② Solar function (the solar thermal panels connected to either the DHW (Domestic Hot Water) Tank or the Buffer Tank.
  • DHW is not applicable for WH-ADC *models.
③ External compressor switch.
④ External error signal.
⑤ SG ready control.
⑥ Demand control.
⑦ Heat-Cool SW

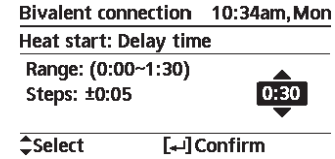
5.2 > Zone & Sensor		
To select the sensors and to select either 1 zone or 2 zone system.	**Zone** • After selecting 1 or 2 zone system, proceed to the selection of room or swimming pool. • If the swimming pool is selected, the temperature must be selected for △T temperature between 0°C ~ 10 °C.	Zone & Sensor 10:34am,Mon Zone 1 Zone system 2 Zones system ▼Select [↵]Confirm
	Sensor * For room thermostat, there is a further selection of external or internal. • If select internal, there is a further selection of RC-1 or RC-2 (only available when Zone selection is 1 zone system). Select RC-1 if main remote controller's thermistor is to be used for room temperature control and vice versa.	Zone & Sensor 10:34am,Mon Sensor Water temperature Room thermostat Room thermistor ▼Select [↵]Confirm
5.3 > Heater capacity		
To reduce the heater power if unnecessary.* 3 kW / 6 kW / 9 kW * Options of kW vary depending on the model.		Heater capacity 10:34am,Mon 3 kW [↵]Confirm
5.4 > Anti freezing		
To activate or deactivate the water freeze prevention when the system is OFF	Yes	Yes ▼ No

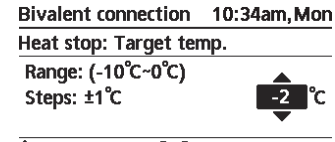
Menu	Default Setting	Setting Options / Display
5.5 > DHW capacity		
To select tank heating capacity to variable or standard. Variable capacity heat up tank with fast mode and keep the tank temperature with efficient mode. While standard capacity heat up tank with rated heating capacity. When “Efficiency” is selected the “5.23 Heating control” setting, the tank heat-up process will be delayed until a much lower tank water temperature is reached, which improves the efficiency of the heating process. To ensure sufficient hot water, select either “Standard” in this setting or “Comfort” in the “5.23 Heating control” setting.	Variable	
5.6 > Buffer tank connection		
To connect tank to the system and if selected YES, to set △T temperature.	No	
	> Yes	
	5 °C	Set △T for Buffer Tank 
5.7 > Base pan heater		
To select whether or not optional base pan heater is connected. * Type A - The base pan heater activates only during deice operation. * Type B - The base pan heater activates when outdoor ambient temperature is 5 °C or lower.	No	
	> Yes	
	A	Set base pan heater type*. 
5.8 > Alternative outdoor sensor		
To select an alternative outdoor sensor.	No	

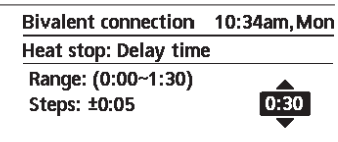
Menu	Default Setting	Setting Options / Display
5.9 > Bivalent connection		
To select to enable or disable bivalent connection.	No	Yes ▲ No
> Yes		
To select either auto control pattern or SG ready input control pattern or smart control pattern. * This selection only display to select when optional pcb connection set to Yes.	Auto	Auto SG ready Smart
To select a bivalent connection to allow an additional heat source such as a boiler to heat-up the buffer tank and domestic hot water tank when heatpump capacity is insufficient at low outdoor temperature. The bivalent feature can be set-up either in alternative mode (heatpump and boiler operate alternately), or in parallel mode (both heatpump and boiler operate simultaneously), or in advance parallel mode (heatpump operates and boiler turns on for buffer-tank and/or domestic hot water depending on the control pattern setting options).	**> Yes > Auto**	
	-5 °C Set outdoor temperature for turn ON Bivalent connection.	Bivalent connection 10:34am,Mon Turn ON: Outdoor temp Range: (-15℃~35℃) Steps: ±1℃ -5 ℃ ▲▼Select [↵]Confirm
	Yes > After selecting the outdoor temperature	
	Control pattern Alternative / Parallel / Advanced parallel • Select advanced parallel for bivalent use of the tanks.	Bivalent connection 10:34am,Mon Control pattern Alternative Parallel Advanced parallel ▲Select [↵]Confirm
	Control pattern > Alternative	
	OFF Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection.	Bivalent connection 10:34am,Mon External pump ON ▲ OFF ▲Select [↵]Confirm
	Control pattern > Advanced parallel	
	Heat Selection of the tank • "Heat" implies Buffer Tank and "DHW" implies Domestic Hot Water Tank.	Bivalent connection 10:34am,Mon Advanced parallel Heat DHW ▼Select [↵]Confirm
	Control pattern > Advanced parallel > Heat > Yes	
	• Buffer Tank is activated only after selecting "Yes".	Bivalent connection 10:34am,Mon Advanced parallel: Heat Yes No ▼Select [↵]Confirm

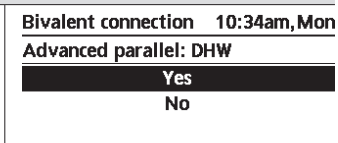
Menu	Default Setting	Setting Options / Display
	-8 °C	Set the temperature threshold to start the bivalent heat source.
	0:30	Delay timer to start the bivalent heat source (in hour and minutes).
	-2 °C	Set the temperature threshold to stop the bivalent heat source.
	0:30	Delay timer to stop the bivalent heat source (in hour and minutes).
	Control pattern > Advanced parallel > DHW > Yes	
	• DHW Tank is activated only after selecting "Yes".	
	0:30	Delay timer to start the bivalent heat source (in hour and minutes).
SG ready input control for bivalent system follow below input condition.	**> Yes > SG ready**	
	OFF	Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection.

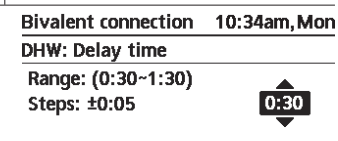




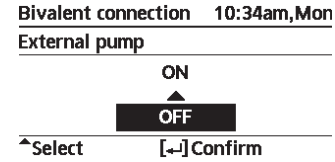




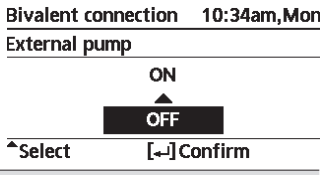


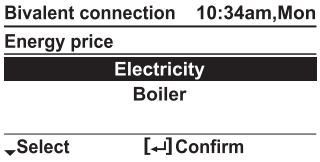


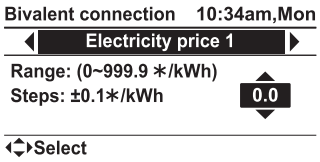
SG signal		Operation pattern
Vcc-bit1	Vcc-bit2	
Open	Open	Heat Pump OFF, Boiler OFF
Short	Open	Heat Pump ON, Boiler OFF
Open	Short	Heat Pump OFF, Boiler ON
Short	Short	Heat Pump ON, Boiler ON

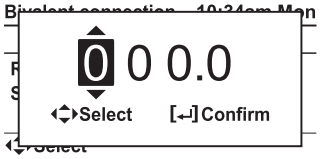


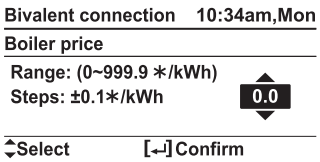
Menu	Default Setting	Setting Options / Display
To do settings related to electricity and boiler so that unit is able to determine whether to operate heat pump or boiler at a particular period depends on operating cost of both heat sources. These settings are electricity price, boiler price, season, schedule etc.	**> Yes > Smart**	
	OFF	Option to set external pump either ON or OFF during bivalent operation. Set to ON if system is simple bivalent connection.
	> Yes > Smart > After selecting for the external pump > Energy price	
	- Select **Electricity** to set on electricity price. - Select **Boiler** to set on boiler price and its efficiency.	
	> Yes > Smart > After selecting for the external pump > Energy price > Electricity	
	0.0 * / kWh - There are total 10 different prices can be set for Electricity: Electricity price 1 ~ Electricity price 10 - Range is 0 ~ 999.9 * / kWh	
	- Press ∧ or ∨ to enter a setting screen as shown in Figure 1. Then start setting the value of electricity price. - After finish setting a particular electricity price (eg. Electricity price 1), press < or > to go and set for other electricity price. * Set the price according to value provided by electrical supply company.	Figure 1
	> Yes > Smart > After selecting for the external pump > Energy price > Boiler	
	0.0 * / kWh - Refer to method of Electricity price setting above for setting of boiler price. - After finish setting of boiler price, set the boiler efficiency (Range : 0 ~ 99%).	
	0% * Set the price according to value provided by boiler or gas supply company.	

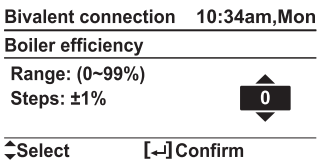




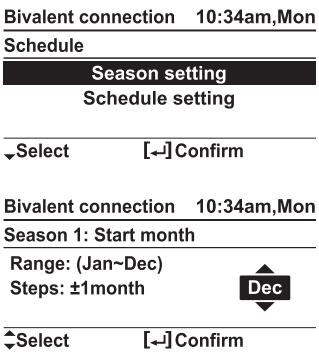
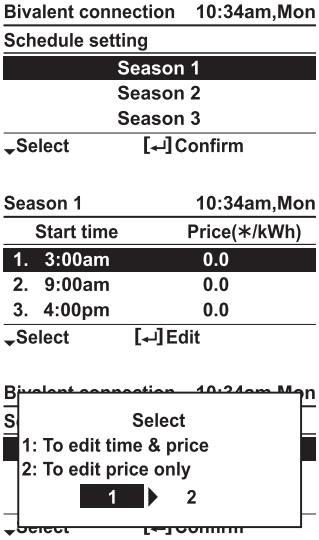


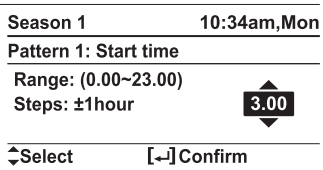
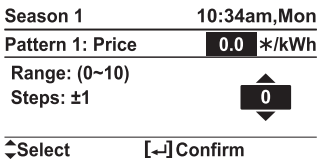
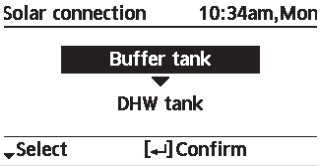
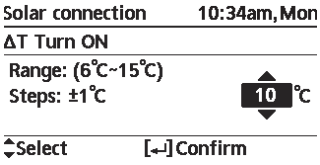


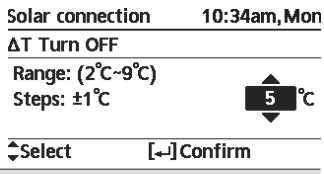
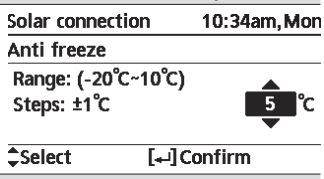
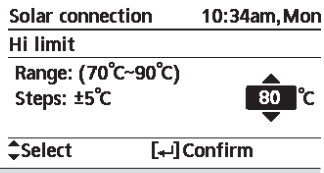
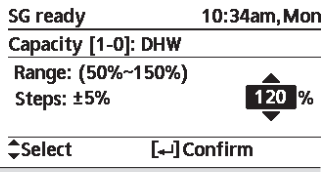
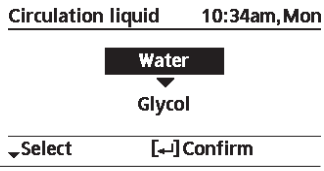




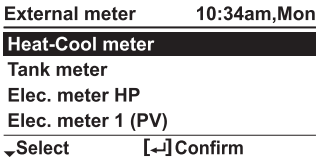
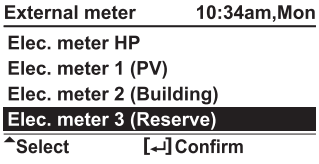
Remark : * implies cents in most currency except Czech crown.

Menu	Default Setting	Setting Options / Display
	> Yes > Smart > After selecting for the external pump > Schedule > Season setting	
	Season 1 : Dec (Refers to Winter season) Season 2 : Mar (Refers to Spring season) Season 3 : Jun (Refers to Summer season) Season 4 : Oct (Refers to Autumn season) - There are total 4 seasons to be set - Set the starting month for each season. (Eg. when Season 1 is set to Dec and Season 2 is set to Mar, month of December to February will be treated as Season 1).	
	> Yes > Smart > After selecting for the external pump > Schedule > Schedule setting	
	Start time (Pattern 1) : 3:00am Start time (Pattern 2) : 9:00am Start time (Pattern 3) : 4:00pm Start time (Pattern 4) : 9:00pm - For each season, there are total 4 patterns can be set. Price (Pattern 1/2/3/4) : 1 - Set the target start time and the appropriate electricity price for each pattern. - Select “1” to edit both start time and electricity price. Select “2” to edit electricity price only.	

Menu	Default Setting	Setting Options / Display	
	- Range of start time displayed can be in “24h” or “am/pm” format depend on setting of “Clock format”.		
	- Range of electricity price is 0 ~ 10 which refers back to the 10 different electricity price set previously (under “Energy price > Electricity”: Electricity price 1 ~ Electricity price 10). The price displayed on the upper right corner indicates the previous set value of Electricity price 1 to Electricity price 10. * When the price is set to “0”, the electricity price will be treated as 0.0 * / kWh. It is for the convenience of installer when 0.0 is the desired setting value for a particular time.		
5.10	**> External SW**		
	No		
5.11	**> Solar connection**		
• The optional PCB connectivity must be selected YES to enable the function. • If the optional PCB connectivity is not selected, the function will not appear on the display. • DHW is not applicable for WH-ADC *models.	No		
	> Yes		
	Buffer tank	Selection of the tank	
	> Yes > After selecting the tank		
	10 °C	Set △T ON temperature	

Menu	Default Setting	Setting Options / Display
	> Yes > After selecting the tank > △T ON temperature	
	5 °C	Set △T OFF temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature	
	5 °C	Set Antifreeze temperature 
	> Yes > After selecting the tank > △T ON temperature > △T OFF temperature > After setting the antifreeze temperature	
	80 °C	Set Hi limit 
5.12 > External error signal		
	No	
5.13 > Demand control		
	No	
5.14 > SG ready		
	No	
	> Yes	
	120 %	Capacity (1) & (2) of DHW (in %), Heat (in %) and Cool (in °C) 
5.15 > External compressor SW		
	No	
5.16 > Circulation liquid		
To select whether to circulate water or glycol in the system.	Water	

Menu	Default Setting	Setting Options / Display
5.17 > Heat-Cool SW		
	No	Yes ▲ No
5.18 > Force heater		
To turn on Force heater either manually (by default) or automatically.	Manual	Force heater 10:34am,Mon Auto ▲ Manual ▲Select [↵] Confirm
5.19 > Force defrost		
If auto selection is set, outdoor unit will start defrost operation if long heating hour operate during low outdoor temperature.	Manual	Auto ▲ Manual
5.20 > Defrost signal		
To turn on defrost signal to stop fan coil during defrost operation. (If defrost signal set to yes, bivalent function will not available to use)	No	Yes ▲ No
5.21 > Pump flowrate		
To set variable flow pump control or fix pump duty control.	△T	ΔT ▼ Max. Duty
5.22 > DHW Defrost		
Allow system to run defrost by using hot water instead of room unit for better room comfort.	Yes	Yes ▼ No
5.23 > Heating control		
To select unit operation condition whether to achieve set temperature faster or to save energy. It is a function for Heating + Tank operation. When “Variable” is selected in the “5.5 DHW capacity” setting, the tank heat-up process will be delayed until a much lower tank water temperature is reached, which improves the efficiency of the heating process. To ensure sufficient hot water, select either “Comfort” in this setting or “Standard” in the “5.5 DHW capacity” setting.	Comfort	Comfort ▼ Efficiency

Menu	Default Setting	Setting Options / Display
5.24 > External meter		
To set which external meter to be used depends on meter connection. There are generation meters and various types of electricity meters. For generation meters, there are two connection systems :- a) One generation meter system : Heat-cool meter only b) Two generation meter system : Heat-cool meter and Tank meter	Heat-cool meter : No * Tank meter : No Elec. meter HP : No Elec. meter 1 (PV) : No Elec. meter 2 (Building) : No Elec. meter 3 (Reserve) : No * Only available when Heat-cool meter select Yes	 
	> Heat-cool meter	
	- Set Heat-cool meter to Yes when this generation meter is connected. - It is to measure energy generation of heat pump unit during heating, cooling and DHW operation (one generation meter system) or during heating and cooling only (two generation meter system).	
	> Tank meter	
	- Set Tank meter to Yes when this generation meter is connected. - It is to measure energy generation of heat pump unit during DHW operation. * Only available to select when Heat-cool meter is set to Yes. Only set Tank meter to Yes when the connection is two generation meter system.	
	> Elec. meter HP	
	- Set Elec. meter HP to Yes when this electricity meter is connected. - It is to measure energy consumption of heat pump unit.	
	> Elec. meter 1 (PV)	
	- Set Elec. meter 1 (PV) to Yes when this electricity meter is connected. - It is to measure energy generation of solar system. * This data will be displayed only on Cloud system.	
	> Elec. meter 2 (Building)	
	- Set Elec. meter 2 (Building) to Yes when this electricity meter is connected. - It is to measure energy consumption of the building. * This data will be displayed only on Cloud system.	

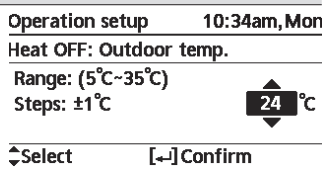
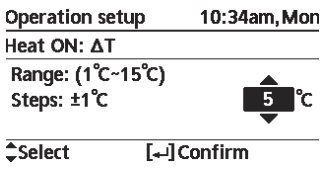
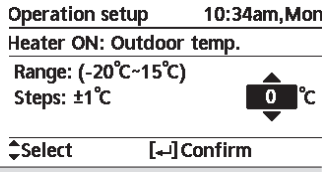
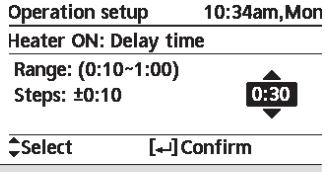
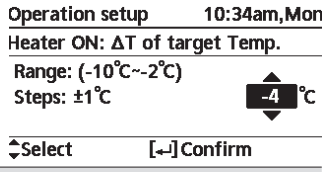
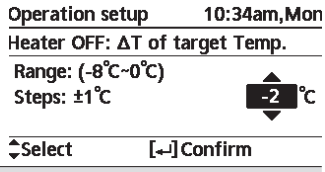
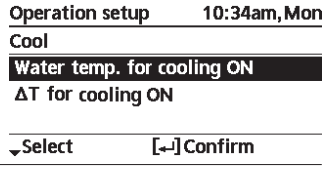
Menu	Default Setting	Setting Options / Display
	> Elec. meter 3 (Reserve)	
	- Set Elec. meter 3 (Reserve) to Yes when this electricity meter is connected. - It is to measure energy consumption * This data will be displayed only on Cloud system.	Yes No
5.25 > Electrical anode		
To enable or disable operation of electrical anode.	Yes (for -AN models) No (for non -AN models)	Yes No

(NOTE) : If [Approx.] is shown on Energy Monitor display, data displayed on the remote controller is obtained through heat pump's internal calculation.
If [Approx.] is NOT shown on Energy Monitor display, data** displayed on the remote controller is obtained by External Meters.
Data stored on the Aquarea unit can be mixed between internal calculation and External Meters.
**In order to know the exact consumption or generation, please use as reference always the External Meters' data.

Remark : Elec. stands for "Electricity"
HP stands for "Heat pump"

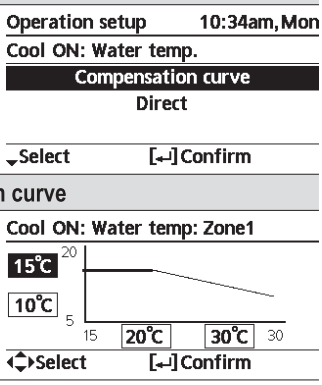
Menu	Default Setting	Setting Options / Display
6 Installer setup > Operation setup		
To access to the four major functions or modes.	4 main modes Heat / *1, *2 Cool / *1, *2 Auto / Tank	Operation setup 10:34am,Mon Heat Cool Auto Tank Select [↵]Confirm
6.1 > Heat		
To set various water & ambient temperatures for heating.	Water temp. for heating ON / Outdoor temp. for heating OFF / △T for heating ON / Heater ON/OFF	Operation setup 10:34am,Mon Heat Water temp. for heating ON Outdoor temp. for heating OFF ΔT for heating ON Select [↵]Confirm
	> Water temp. for heating ON	
	Compensation curve Heating ON temperatures in compensation curve or direct input.	Operation setup 10:34am,Mon Heat ON: Water temp. Compensation curve Direct Select [↵]Confirm
	> Water temp. for heating ON > Compensation curve	
	X axis: -5 °C, 15 °C Y axis: 55 °C, 35 °C Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis).	Heat ON: Water temp.:Zone1 55°C 75 35°C 20 -20 -5°C 15°C 15 Select [↵]Confirm
	• Temperature range: X axis: -20 °C ~ 15 °C, Y axis: See below • Temperature range for the Y axis input: WH-WDG model: 20 °C ~ 75 °C Regardless of the above setting, there is a limit to the water temperature. Refer to the operating condition on page 3. • If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> Water temp. for heating ON > Direct	
	35 °C Temperature for heating ON	Operation setup 10:34am,Mon Heat ON: Water temp.:Zone2 Range: (20°C~75°C) Steps: ±1°C 35 °C Select [↵]Confirm
	• Min. ~ Max. range is 20 °C ~ 75 °C: WH-WDG model: 20 °C ~ 75 °C Regardless of the above setting, there is a limit to the water temperature. Refer to the operating condition on page 3. • If 2 zone system is selected, temperature set point must input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	

*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Outdoor temp. for heating OFF	
	24 °C	Temperature for heating OFF 
	> △T for heating ON	
	5 °C	Set △T for heating ON. * This setting will not available to set when pump flowrate set to Max. duty. 
	> Heater ON/OFF	
	> Heater ON/OFF > Outdoor temp. for heater ON	
	0 °C	Temperature for heater ON 
	> Heater ON/OFF > Delay time for heater ON	
	0:30 min	Delay time for heater to turn on 
	> Heater ON/OFF > Water temperature for heater ON	
	-4 °C	Setting of water temperature to turn on from water set temperature. 
	> Heater ON/OFF > Water temperature for heater OFF	
	-2 °C	Setting of water temperature to turn off from water set temperature. 
6.2	**> *1, *2 Cool**	
To set various water & ambient temperatures for cooling.	Water temperatures for cooling ON and △T for cooling ON.	

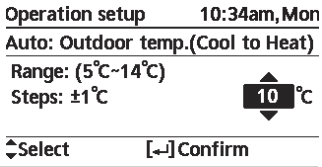
*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

Menu	Default Setting	Setting Options / Display
	> Water temp. for cooling ON	
	Compensation curve	Cooling ON temperatures in compensation curve or direct input.
	> Water temp. for cooling ON > Compensation curve	
	X axis: 20 °C, 30 °C Y axis: 15 °C, 10 °C	Input the 4 temperature points (2 on horizontal X axis, 2 on vertical Y axis)
	• If 2 zone system is selected, the 4 temperature points must also be input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> Water temp. for cooling ON > Direct	
	10 °C	Set temperature for Cooling ON
	• If 2 zone system is selected, temperature set point must input for Zone 2. • "Zone 1" and "Zone 2" will not appear on the display if only 1 zone system.	
	> △T for cooling ON	
	5 °C	Set △T for cooling ON * This setting will not available to set when pump flowrate set to Max. duty.
6.3	**> *1, *2 Auto**	
Automatic switch from Heat to Cool or Cool to Heat.	Outdoor temperatures for switching from Heat to Cool or Cool to Heat. Outdoor temp. for (Heat to Cool) / Outdoor temp. for (Cool to Heat)	
	> Outdoor temp. for (Heat to Cool)	
	15 °C	Set outdoor temperature for switching from Heat to Cool.



*1 The system is locked to operate without COOL mode. It can be unlocked only by authorised installers or our authorised service partners.
*2 Only displayed when COOL mode is unlocked (This means when COOL mode is available).

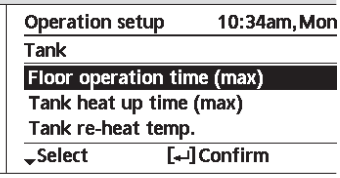
Menu	Default Setting	Setting Options / Display
	> Outdoor temp. for (Cool to Heat)	
	10 °C	Set outdoor temperature for switching from Cool to Heat.



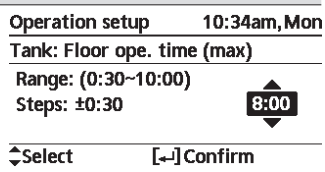
6.4 > Tank

Setting functions for the tank.

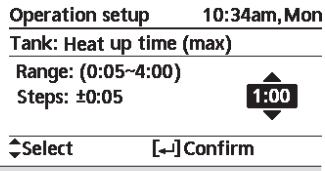
Floor operation time (max) / Tank heat up time (max) / Tank re-heat temp. / Sterilization	• The display will show 3 functions at a time.



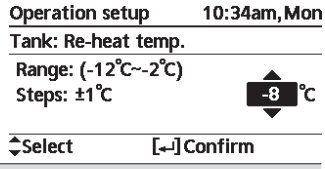
> Floor operation time (max)		
8:00	Maximum time for floor operation (in hours and minutes)	



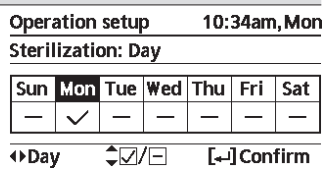
> Tank heat up time (max)		
1:00	Maximum time for heating the tank (in hours and minutes)	



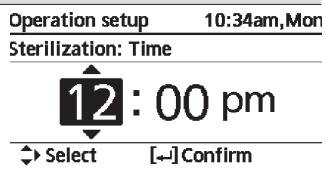
> Tank re-heat temp.		
-8 °C	Set temperature to perform reboil of tank water.	



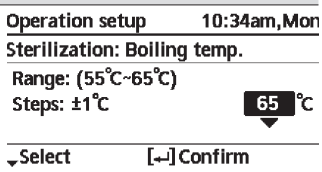
> Sterilization		
Monday	Sterilization may be set for 1 or more days of the week. Sun / Mon / Tue / Wed / Thu / Fri / Sat	

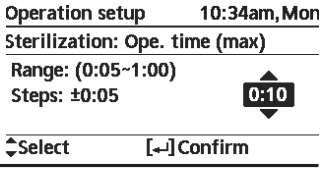


> Sterilization: Time		
12:00	Time of the selected day(s) of the week to sterilize the tank 0:00 ~ 23:59	



Menu	Default Setting	Setting Options / Display
	> Sterilization: Boiling temp.	
	65 °C	Set boiling temperatures for sterilize the tank.
	> Sterilization: Ope. time (max)	
	0:10	Set sterilizing time (in hours and minutes)





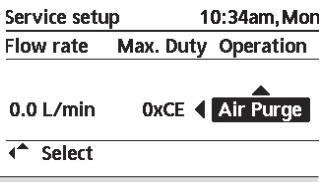
7 Installer setup > Service setup

7.1 > Pump maximum speed

To set the maximum speed of the pump.

Setting the flow rate, max. duty and operation ON/OFF of the pump.

Flow rate: XX:X L/min
Max. Duty: 0x40 ~ 0xFE,
Pump: ON/OFF/Air Purge



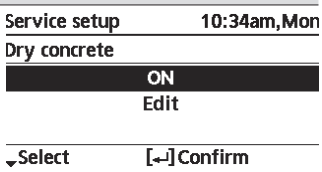
7.2 > Dry concrete

To dry the concrete (floor, walls, etc.) during construction.

Do not use this menu for any other purposes and in period other than during construction

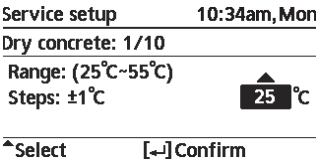
Edit to set the temperature of dry concrete.

ON / Edit



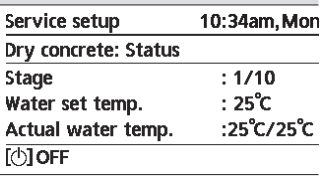
> Edit

Stages: 1 Temperature: 25 °C	Heating temperature for drying the concrete. Select the desired stages: 1 ~ 10, range: 1 ~ 99



> ON

Confirm the setting temperatures of dry concrete for each stage.

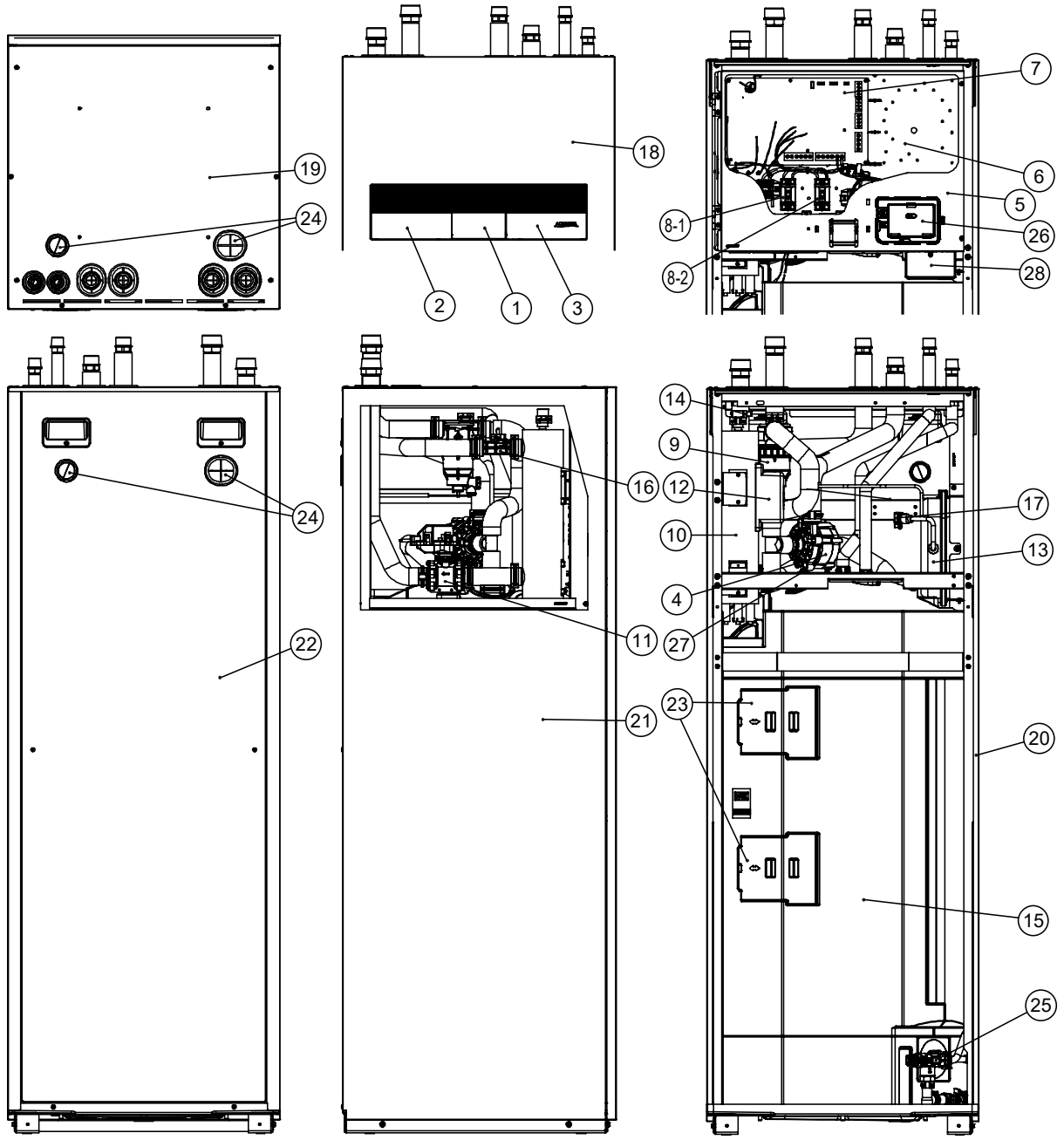


Menu	Default Setting	Setting Options / Display
7.3 > Service contact		
To set up to 2 contact names and numbers for the User.	Service engineer's name and contact number. Contact 1 / Contact 2	Service setup 10:34am, Mon Service contact: Contact 1 Contact 2 Select [↵] Confirm
	> Contact 1 / Contact 2	
	Contact name or number. Name / phone icon	Service contact 10:34am, Mon Contact 1 Name : Bryan Adams : 08812345678 Select [↵] Edit
	Input name and number Contact name: alphabet a ~ z. Contact number: 1 ~ 9	Contact-1 ABC/abc 0-9/Other ABCDEFGHIJKLMNOPQR Space STUVWXYZ abcdefghi BS jklmnopqrstuvwxyz Conf Select [↵] Enter Number: 1 2 3 (4 5 6) 7 8 9 - BS * 0 # _ Conf Select [↵] Enter

8 Installer setup > Remote control setup			
• To select whether to use one remote controller or two remote controllers. • Select Single when one remote controller is connected. Select Dual when two remote controllers are connected. Second remote controller can be used for zone 2 room temperature control.	Single	Selection of one or two remote controllers.	Single Dual
		When Dual is selected, Main remote controller (RC-1) will start to communicate with second remote controller (RC-2) and display "RC-1 & RC-2 sync. in progress". They are ready to be used after this pop up screen disappears.	RC-1 & RC-2 sync. in progress!
		When both remote controllers have communication failure, it will display "Communication with RC-2 failed".	Communication with RC-2 failed! [⮌] Close

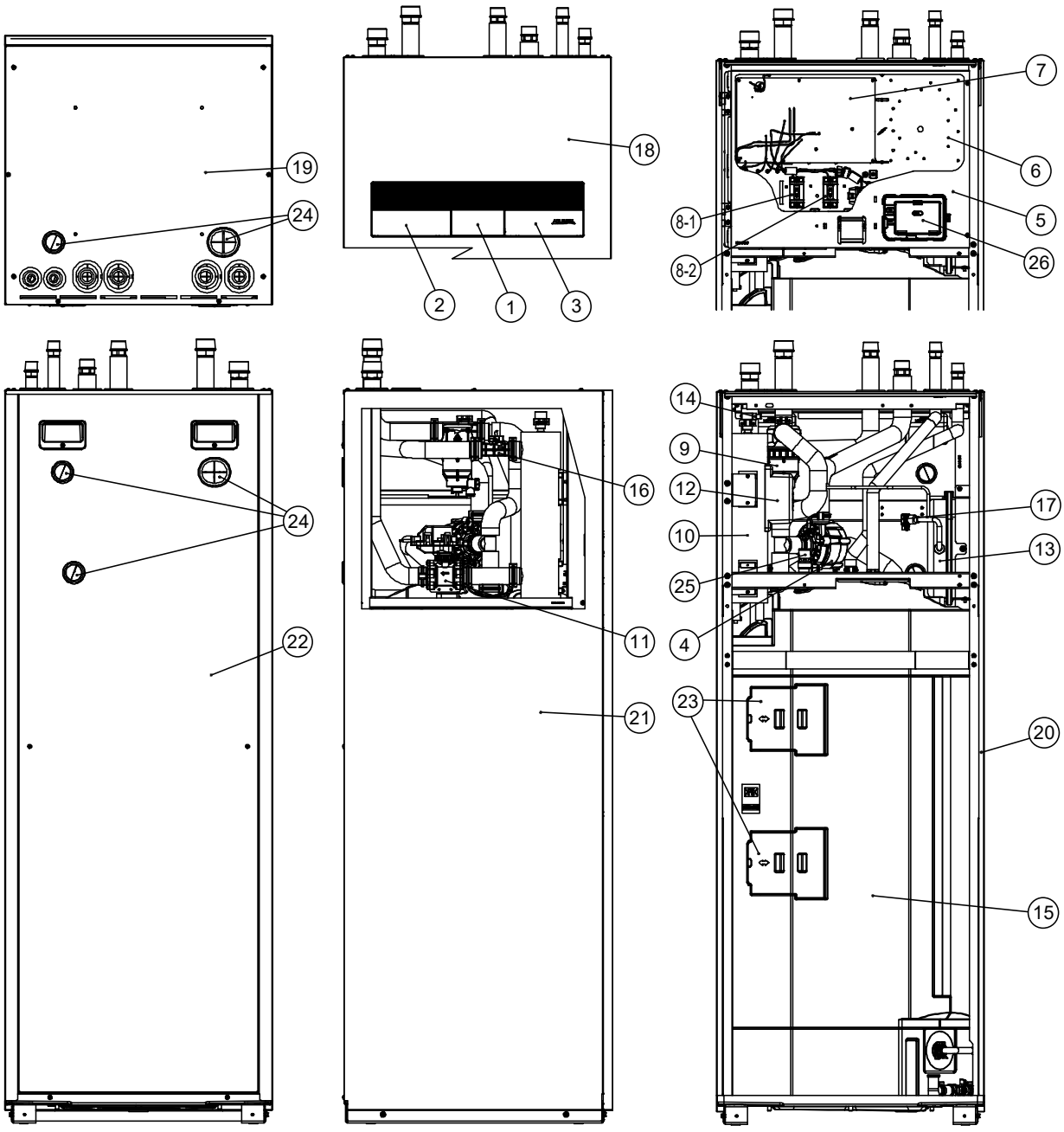
5.1.7 Main Components

5.1.7.1 WH-ADC0509L3E51



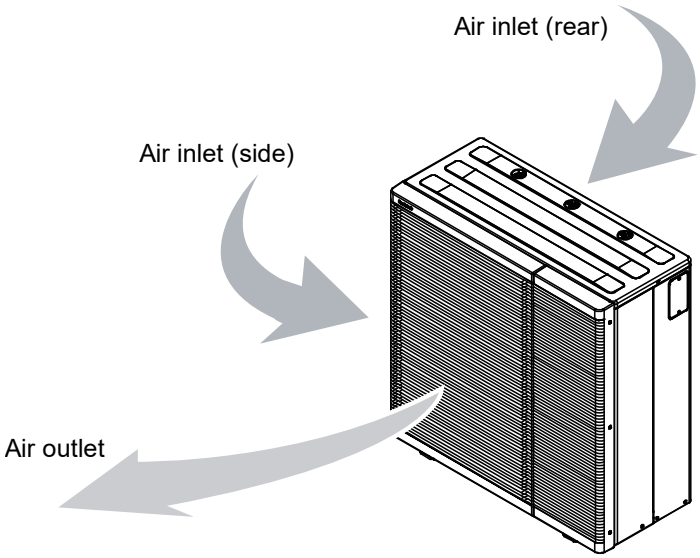
1	Remote Controller	10	Heater Assembly	20	Right Plate
2	Left Decoration Panel	11	3-Way Valve	21	Left Plate
3	Right Decoration Panel	12	Overload Protector (Not Visible)	22	Rear Plate
4	Water Pump	13	Expansion Vessel	23	Tank Sensor (Not Visible)
5	Control Board Cover	14	Drain plug	24	Bushing (4 pieces)
6	Control Board	15	Tank	25	Safety Relief Valve
7	Main PCB	16	Flow Sensor	26	Network Adaptor Holder
8-1	Single Phase RCCB/ELCB (Main Power)	17	Water Pressure Sensor	27	Electric Anode Bar (Not Visible - Applicable only for WH-ADC0509L3E5AN1)
8-2	Single Phase RCCB/ELCB (Backup Heater)	18	Front Plate	28	Electric anode PCB (Not Visible - Applicable only for WH-ADC0509L3E5AN1)
9	Magnetic Water Filter Set	19	Top Plate		

5.1.7.2 WH-ADC0509L3E5UK1



(1)	Remote Controller	(10)	Heater Assembly	(20)	Right Plate
(2)	Left Decoration Panel	(11)	3-Way Valve	(21)	Left Plate
(3)	Right Decoration Panel	(12)	Overload Protector (Not Visible)	(22)	Rear Plate
(4)	Water Pump	(13)	Expansion Vessel	(23)	Tank Sensor (Not Visible)
(5)	Control Board Cover	(14)	Drain plug	(24)	Bushing (5 pieces)
(6)	Control Board	(15)	Tank	(25)	Temperature and Pressure Relief Valve
(7)	Main PCB	(16)	Flow Sensor	(26)	Network Adaptor Holder
(8-1)	Single Phase RCCB/ELCB (Main Power)	(17)	Water Pressure Sensor		
(8-2)	Single Phase RCCB/ELCB (Backup Heater)	(18)	Front Plate		
(9)	Magnetic Water Filter Set	(19)	Top Plate		

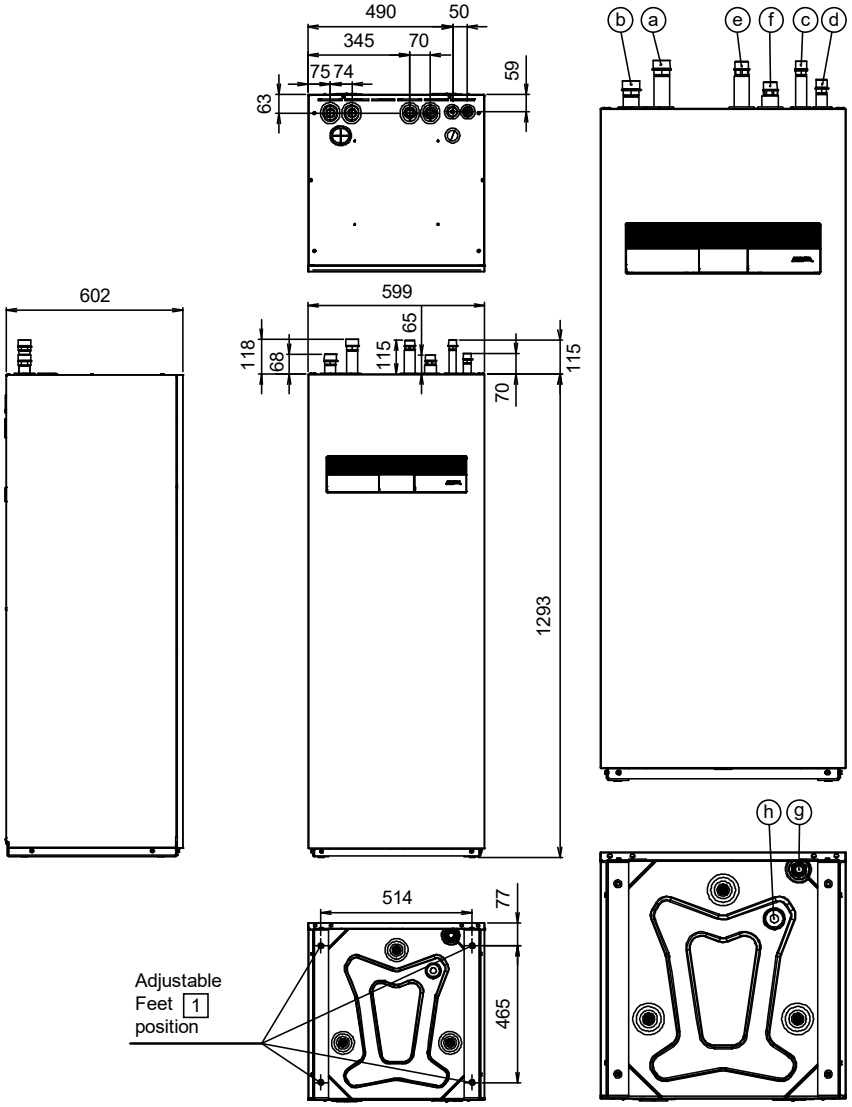
5.2 Outdoor Unit



6. Dimensions

6.1 Indoor Unit

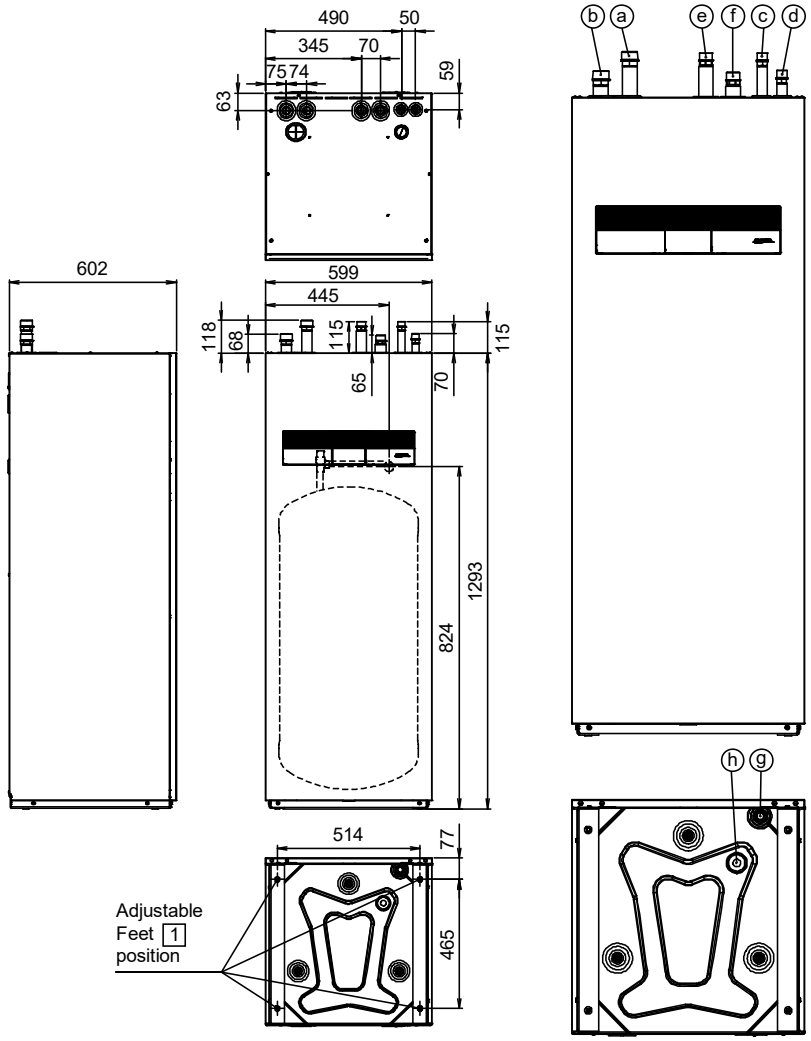
6.1.1 WH-ADC0509L3E51



Tube Connector	Function	Connector Size
ⓐ	Zone 1 Water Inlet (From Space Heating/Cooling)	R 1¼"
ⓑ	Zone 1 Water Outlet (To Space Heating/Cooling)	R 1¼"
ⓒ	Cold Water Inlet (Domestic Hot Water Tank)	R ¾"
ⓓ	Hot Water Outlet (Domestic Hot Water Tank)	R ¾"
ⓔ	Water Inlet (From Outdoor Unit)	R 1"
ⓕ	Water Outlet (To Outdoor Unit)	R 1"
ⓖ	Domestic Hot Water Tank Discharge (Drain Tap) Type: Ball Valve	Rc 1/2"
ⓗ	Drain Water Hole	---

Unit: mm

6.1.2 WH-ADC0509L3E5UK1

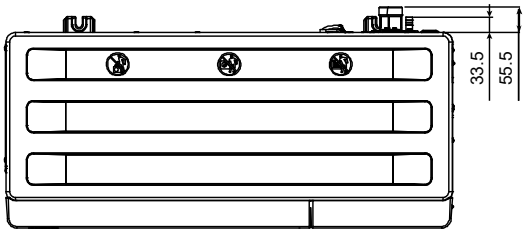


Tube Connector	Function	Connector Size
ⓐ	Zone 1 Water Inlet (From Space Heating/Cooling)	R 1¼"
ⓑ	Zone 1 Water Outlet (To Space Heating/Cooling)	R 1¼"
ⓒ	Cold Water Inlet (Domestic Hot Water Tank)	R ¾"
ⓓ	Hot Water Outlet (Domestic Hot Water Tank)	R ¾"
ⓔ	Water inlet (From Outdoor Unit)	R 1"
ⓕ	Water outlet (To Outdoor Unit)	R 1"
ⓖ	Domestic Hot Water Tank Discharge (Drain Tap) Type: Ball Valve	Rc 1/2"
ⓗ	Drain Water Hole	---

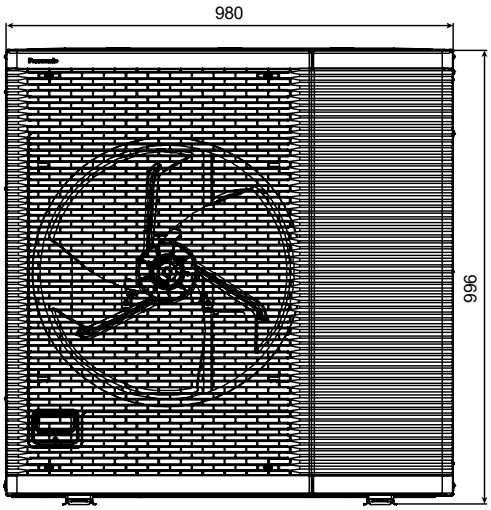
Unit: mm

6.2 Outdoor Unit

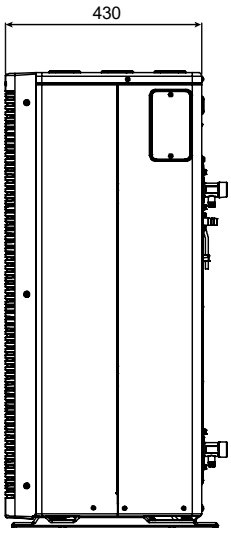
<Top View>



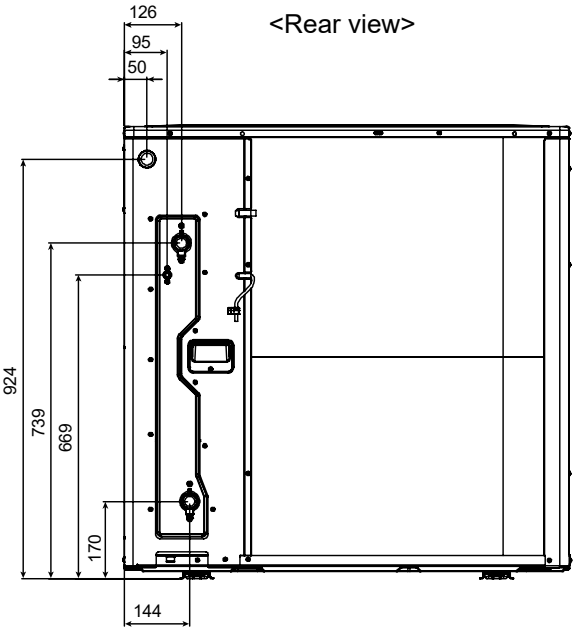
<Front View>



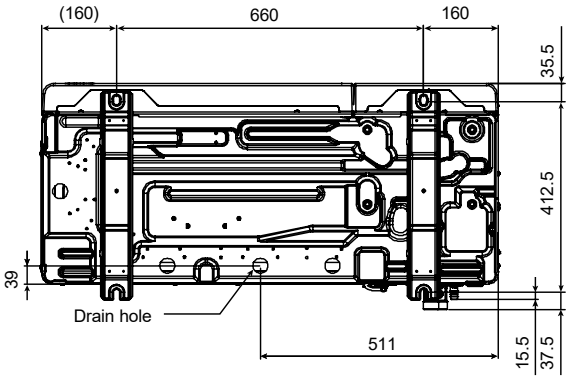
<Side View>



<Rear view>



<Back view>



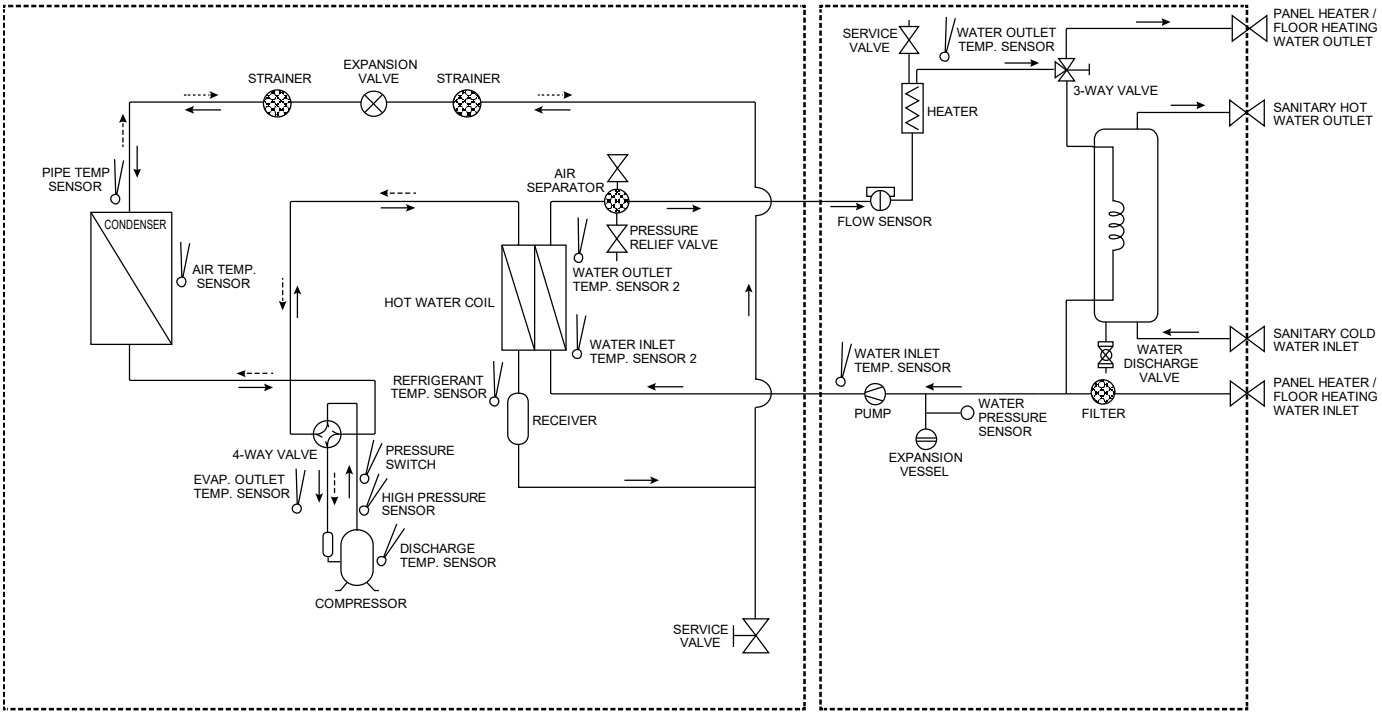
Unit: mm

7. Refrigeration and Water Cycle Diagram

1 ZONE

OUTDOOR UNIT

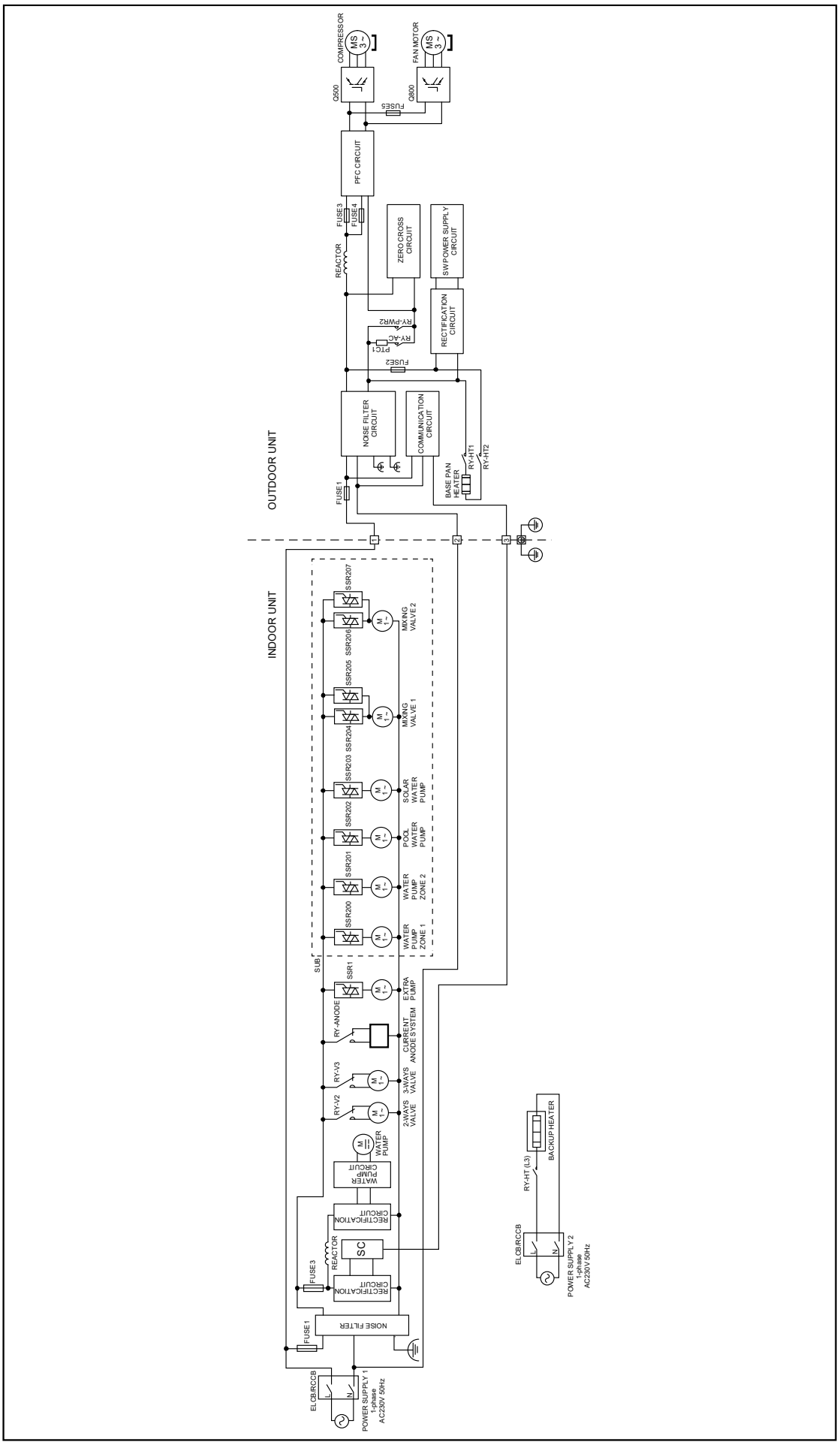
INDOOR UNIT



REFRIGERANT CYCLE [HEATING]

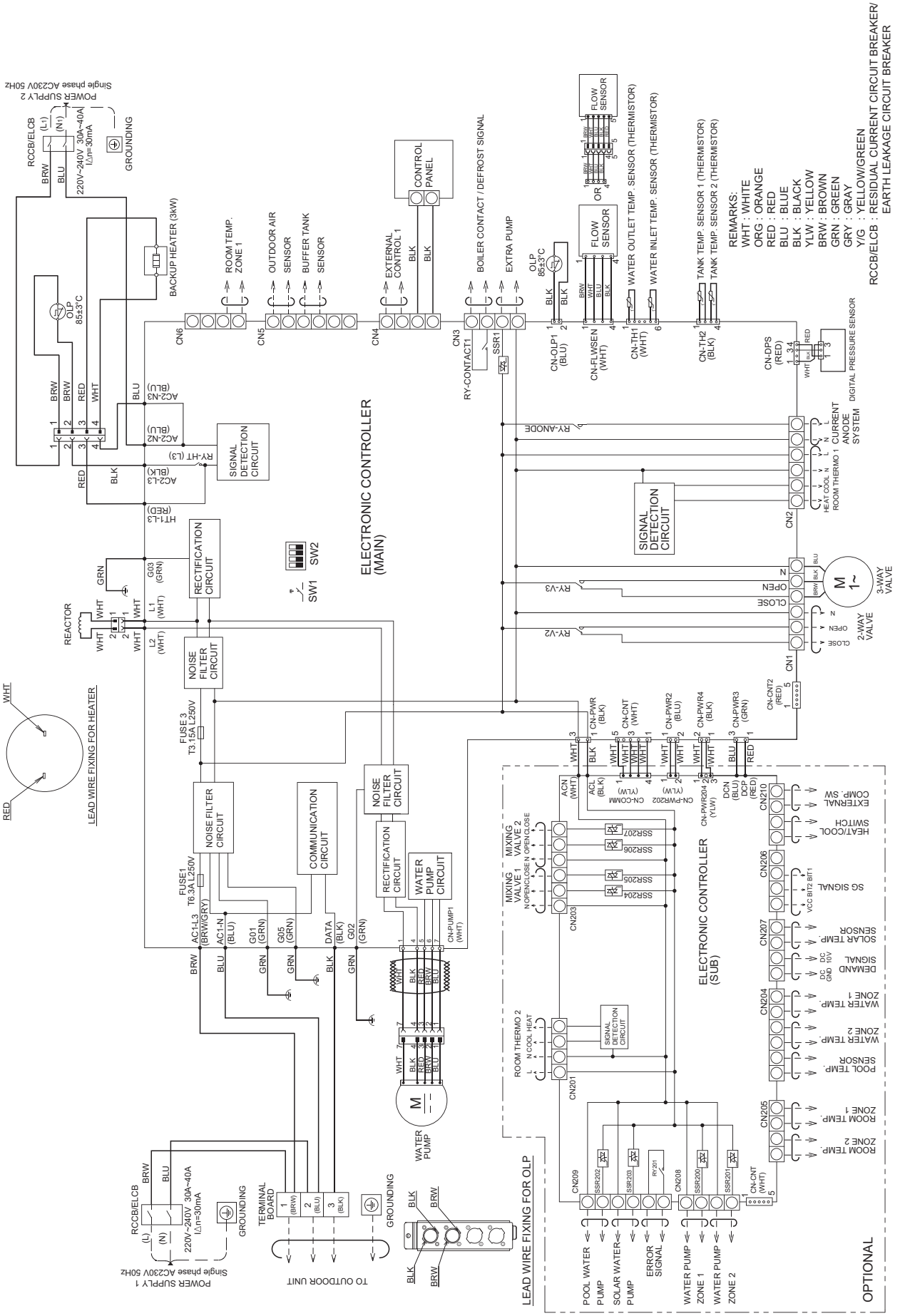
REFRIGERANT CYCLE [COOLING]

8. Block Diagram

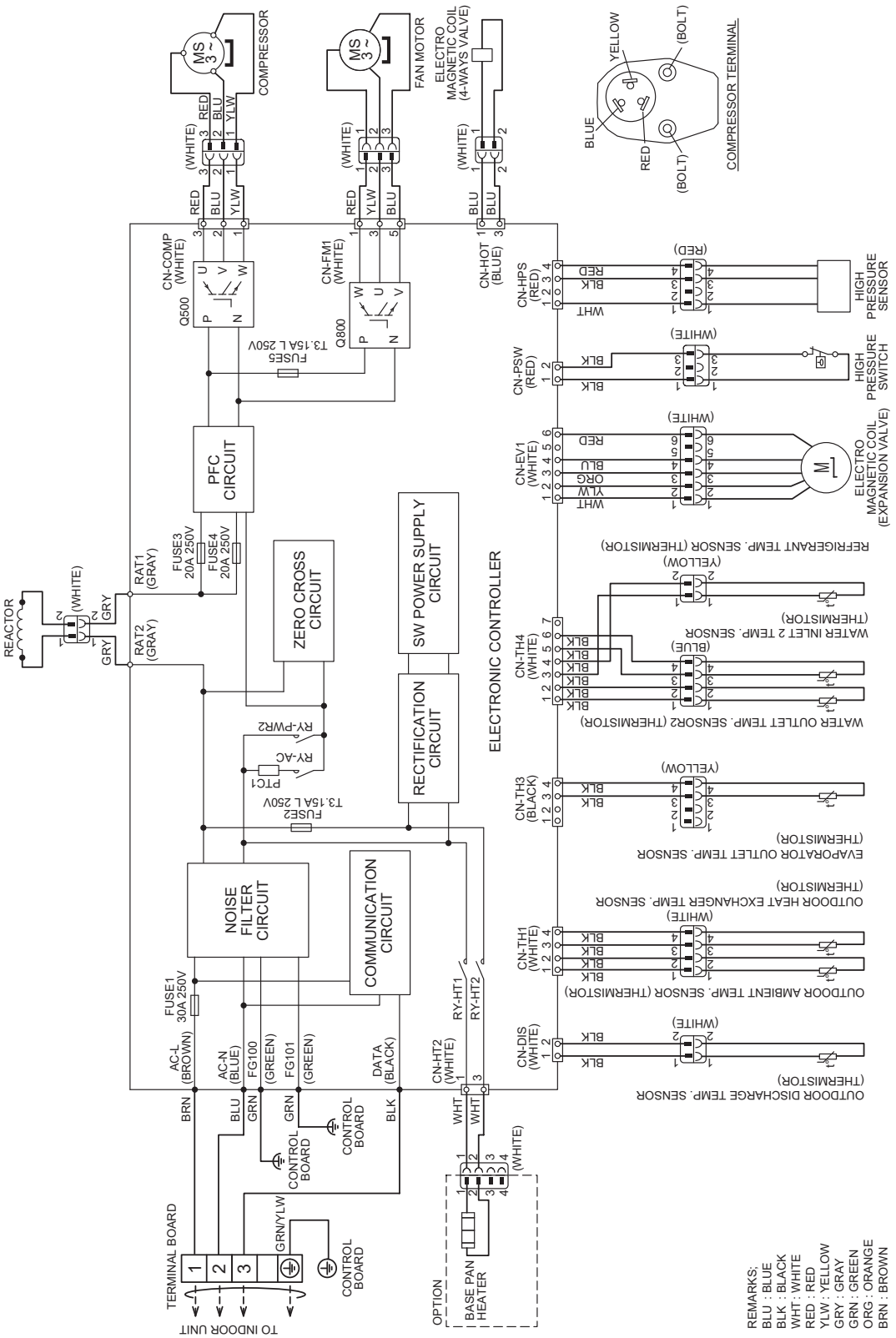


9. Wiring Connection Diagram

9.1 Indoor Unit



9.2 Outdoor Unit



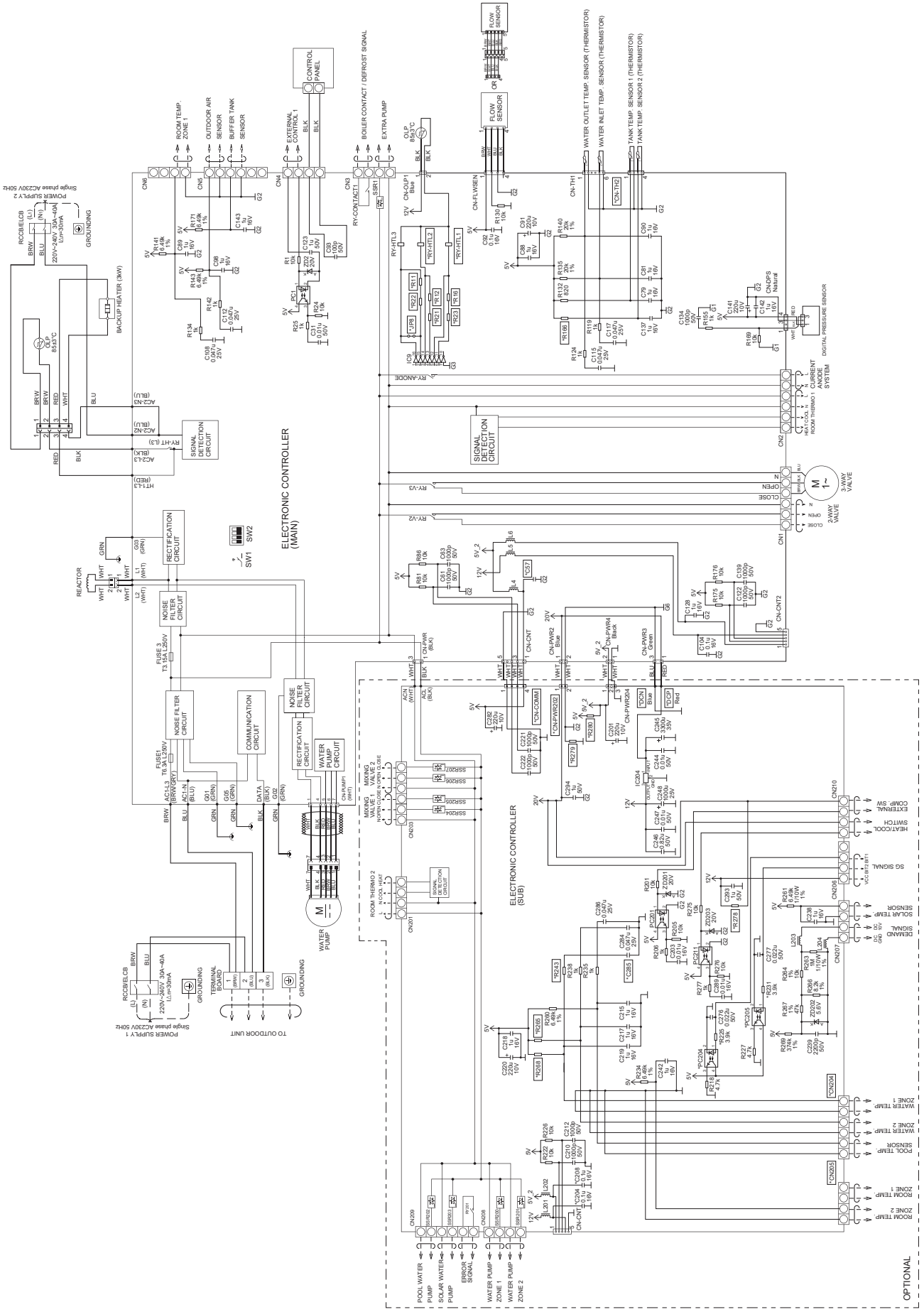
Resistance of Compressor Windings

MODEL	WH-WDG05LE5 / WH-WDG07LE5 / WH-WDG09LE5
CONNECTION	9KD420XAD21
U - V	0.722 Ω
V - W	0.707 Ω
U - W	0.708 Ω

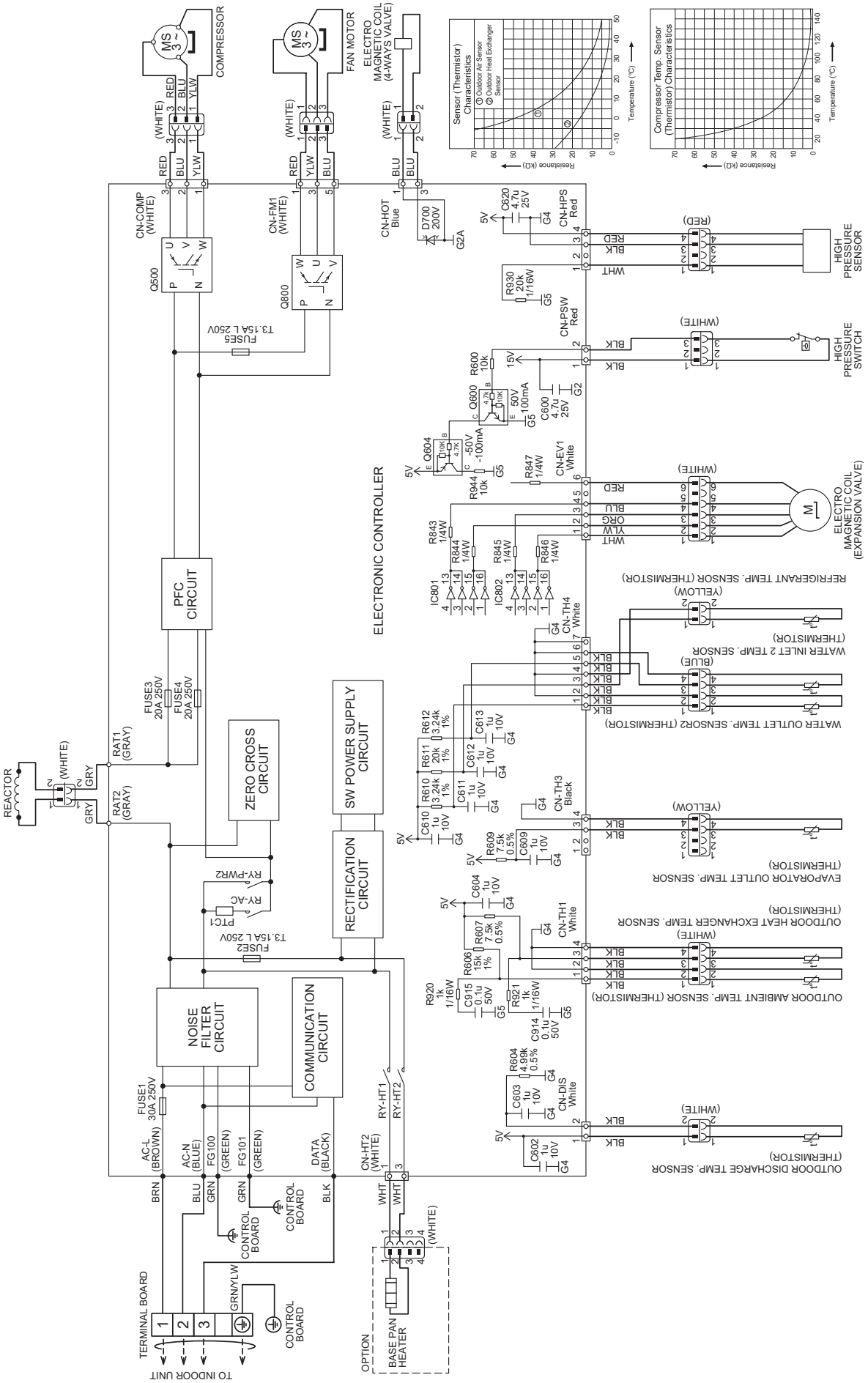
Note: Resistance at 20°C of ambient temperature.

10. Electronic Circuit Diagram

10.1 Indoor Unit



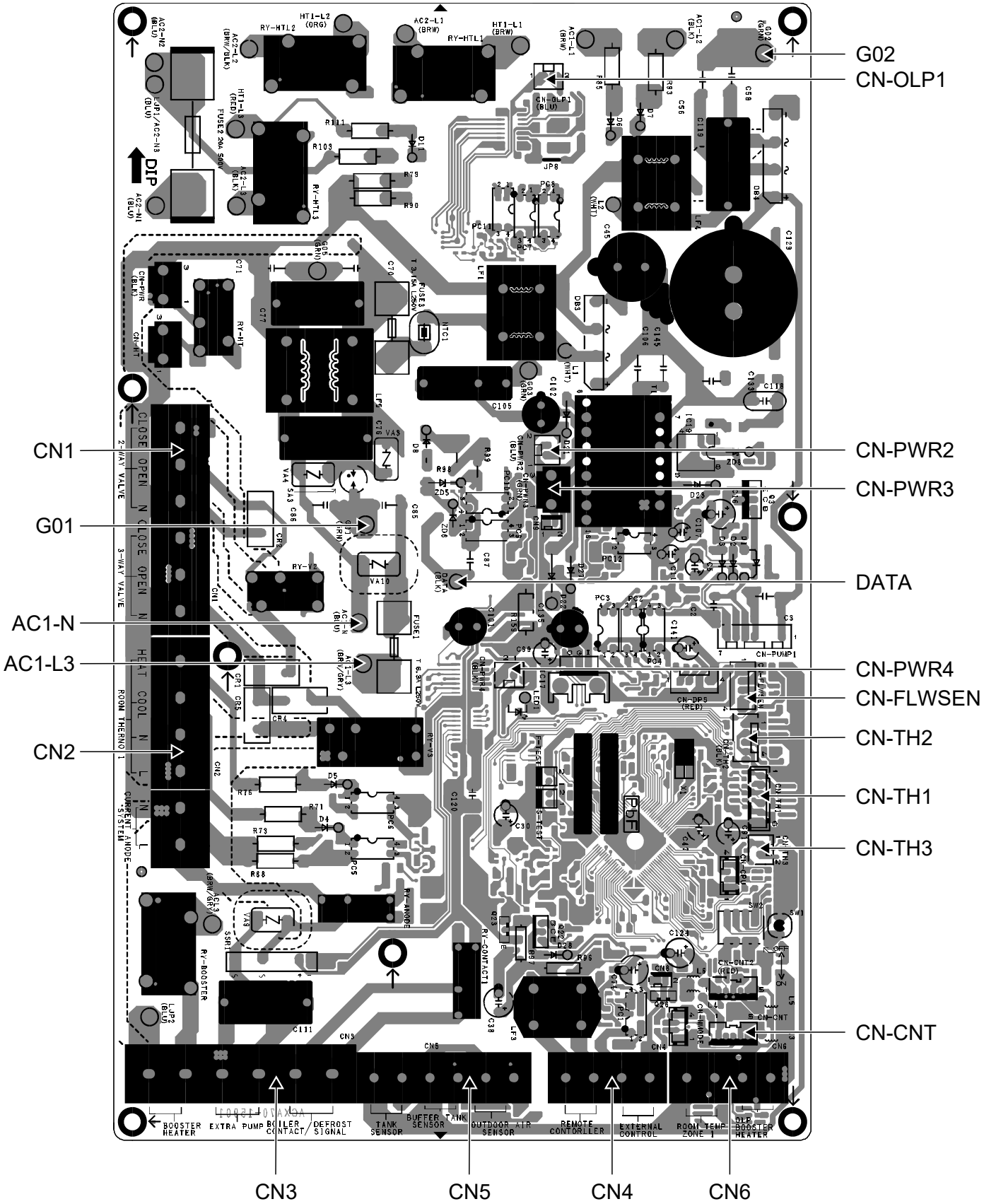
10.2 Outdoor Unit



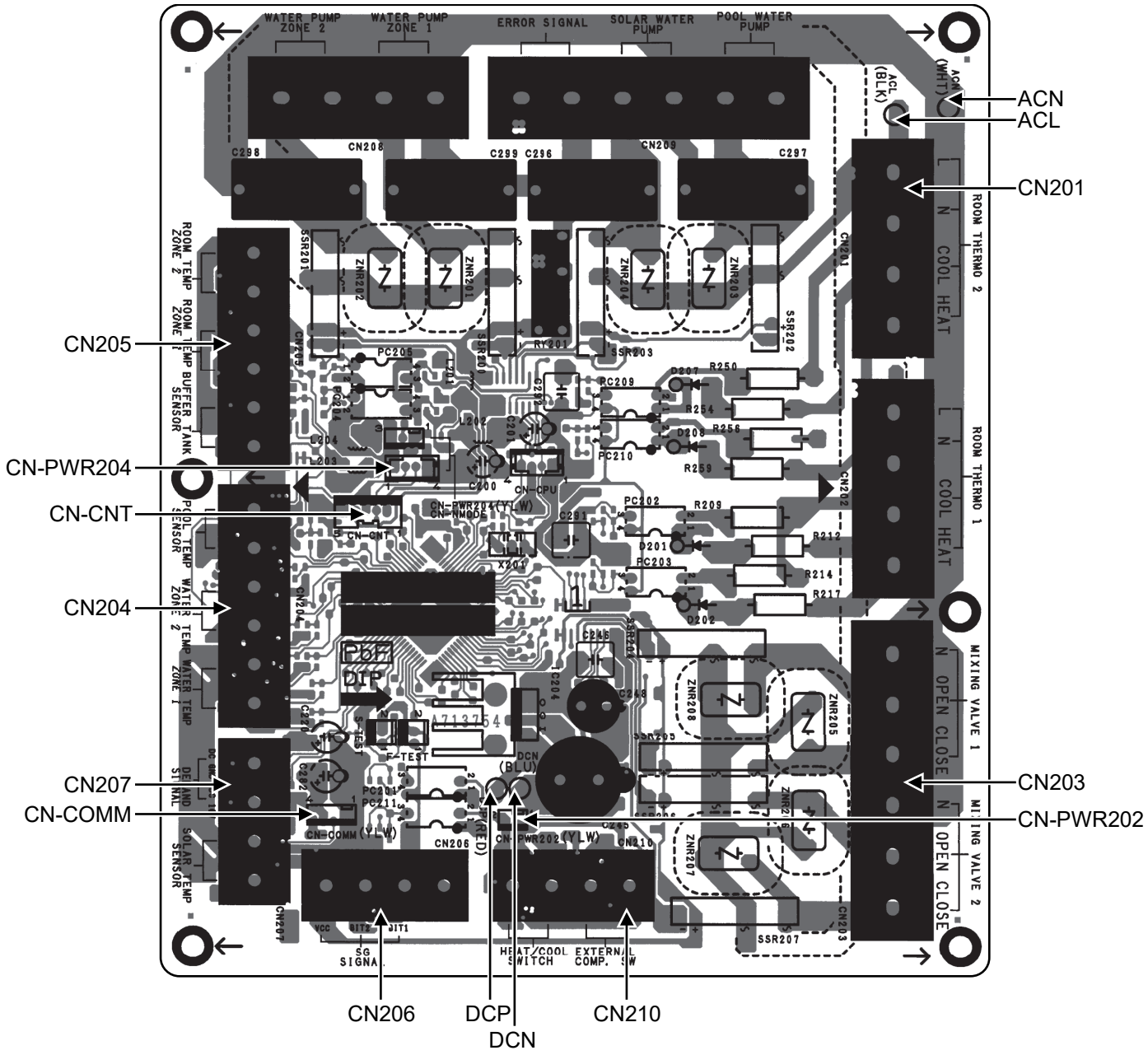
11. Printed Circuit Board

11.1 Indoor Unit

11.1.1 Main Printed Circuit Board



11.1.2 Sub Printed Circuit Board (Optional)



11.2 Outdoor Unit

11.2.1 Main Printed Circuit Board

